

Process Update & Roadmap

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10 May 2017



Agenda

- GaN update in general
- Status of CMOS fab transfer
- Epi update
- Process Corner lots
- Reliability update – bHAST+
- Idq Drift
- Epiwafer technology development lab tour

Agenda

- GaN update in general
 - Why choose *GaN-on-Silicon* ?
 - GaN-on-Silicon Process Technology History
 - Technology Development Roadmaps

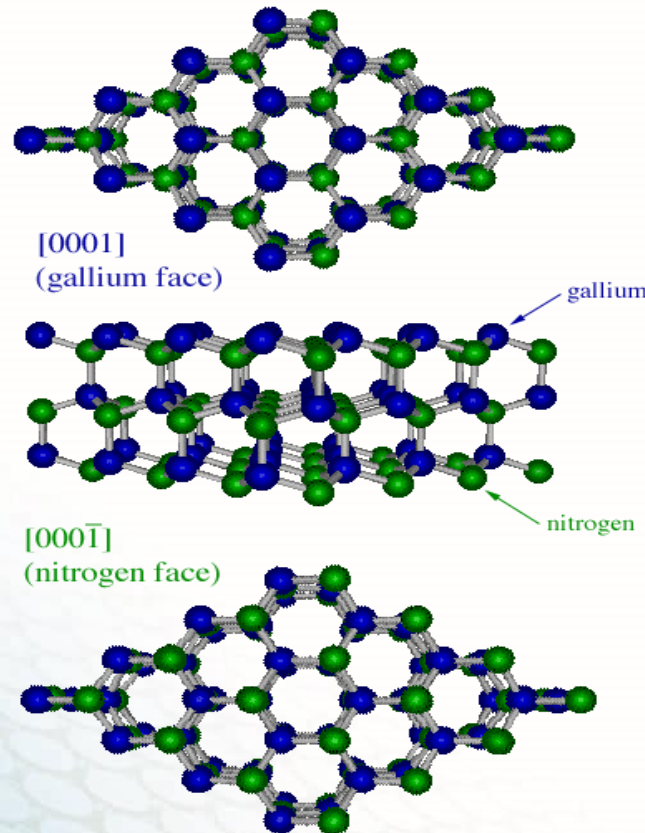
Why Chose “*GaN-on-Silicon*” ?



GaN Heteroepitaxy Substrate Choices

Considerations

- Lattice Parameter
- Thermal Expansion
- Thermal Conductivity
- Microwave Isolation
- **Substrate Cost**
- **Wafer Diameter**
- **Maturity / Quality**
- **Secure Supply**



Substrates Used

- Bulk GaN or AlN (?)
- Sapphire
- **Silicon Carbide**
- **Silicon**
- Zinc Oxide
- LiGaO

Volume Production of Silicon Vs. SiC Ingots used for Wafering Substrates



5) PROCESS TIME FOR FZ CRYSTAL

Q: How long does it take to grow a FZ Si boule?, how long for a CZ Si boule? (will diameter also be a factor here?)

A: Typical timing below table. One FZ crystal pr. day pr. FZ crystal puller. Topsil capacity is 14 pullers, and capacity can easily be expanded by adding more FZ pullers.

Parameter	FZ	CZ	SiC
Crystal pulling speed	4" 2,8 mm/minutes 6" 2,4 mm/minutes 200 mm 2,2 mm/minutes	4" 1 mm/minutes 6" 0,7 mm/minutes 200 mm 0,7 mm/minutes	0,001 mm/minutes
Turnaround time typical for one crystal	One day	Two days	20 days
Process temperature	1420 deg C.	1420 deg C.	2000 - 2500 deg C
Energy consumption	Low	Medium	Very high
Consumables	Polysilicon	Polysilicon + crucible	Polysilicon + carbon + crucible
Maximum diameter	200 mm	300mm	150mm



Confidential

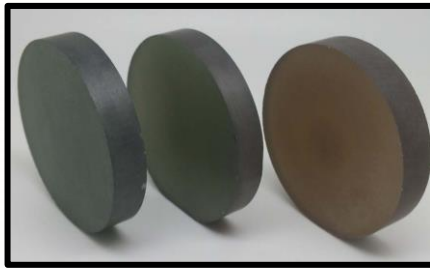
TOPSIL

It takes ~ **1 day** to pull a **4 meter** long **Silicon** Ingot, whereas it takes ~**20 days** to grow a **3 centimeter** long **SiC** ingot.

SiC Substrate Supply Chain Limitations



- Shown Below, Typical SiC Ingots prior to slicing into substrates.
- SiC growth technology (Physical Vapor Transport)
limits Ingot lengths to centimeters and results in minimal wafers per Ingot



100-mm SiC Ingots



50-mm, 75-mm, 100-mm
& 150-mm SiC Ingots

- Questionable SiC Substrate Manufacturing Supply Options

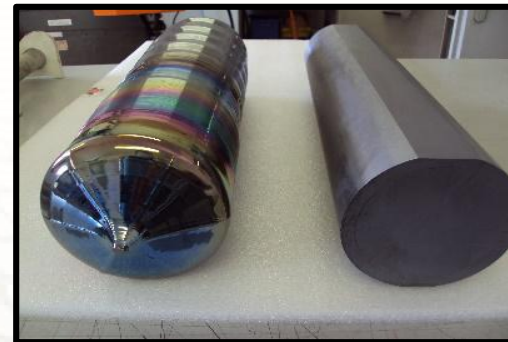
Supplier	Location	100-mm resistive RF 4H-SiC	150-mm resistive RF 4H-SiC
Cree	USA	●	●
II-VI	USA	●	●
Norstel	Sweden	●	●
SiCrystal	Germany	●	●
TanKebblue	China	●	●
Dow Corning	USA	●	●
PAM-XIAMEN	China	●	●
Showa Denko	Japan	?	?
NSSMC	Japan	?	?

*Survey conducted
October 2016*

MACOM Proprietary, Shared with Nokia on 10 May 2017 under strict NDA

Secure Silicon Substrate Supply Chain **MACOM**

- Typical Silicon Ingots can be 2 meters or more in length supporting **thousands of substrates per ingot**, and scales to 300-mm+ diameters.
- **GaN on Silicon** uses the **same high purity polysilicon** starting material and crystal growth techniques to pull silicon ingots from a melt **as LDMOS** and IGBTs, **very mature and reliable technology, secure supply chain.**



MACOM Proprietary, Shared with Nokia on 10 May 2017 under strict NDA

Silicon Vs. SiC Substrate Manufacturing Readiness for the BTS Market Ramp



100's of small SiC ingots =



=

1 Silicon Ingot.



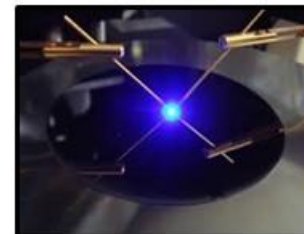
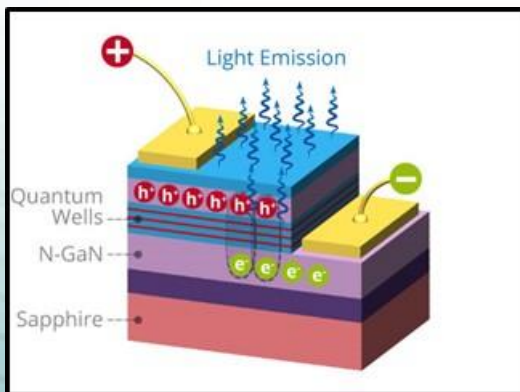
- High volume foundries consume 10,000 wafers/day!



A Word about GaN-based Epiwafer Volume



- The success of the blue LED and solid State Lighting Industry today is based primarily on GaN-on-sapphire and GaN-on-SiC which use wide bandgap substrates to aid in high efficacy blue light emission.
- GaN-on-Silicon was not pursued originally as the silicon substrates (narrow bandgap) absorbed 40% of the possible blue light emission, therefore less efficient.



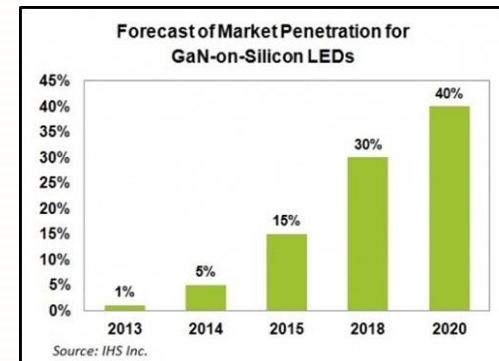
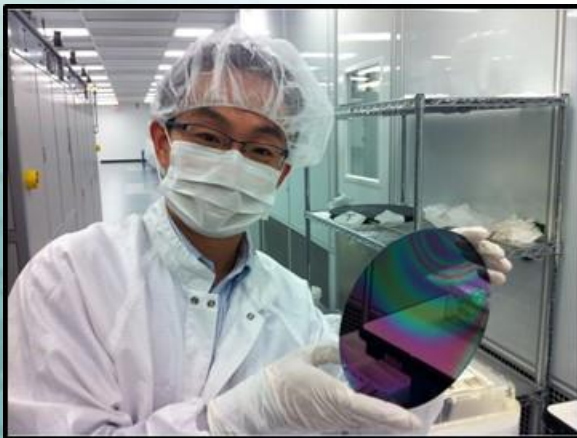
In 2013, GaN LED production exceeded **100 billion units**
- equivalent to 15 LEDs for every person on the planet.

Now Here Come GaN-on-Si LEDs



- Today, LED manufactures are removing the SiC and Sapphire substrates to increase light extraction.
- Cost is driving the industry now over to silicon now that the substrate is being removed.
- Firms who are making LEDs the key focus of GaN-on-silicon include Osram, Samsung, Toshiba and Plessey.

**200-mm GaN-on-Si
Toshiba LED EPIwafer**



GaN-on-Si Street Lights



GaN-on-Si Lightbulbs



And What About GaN-based High Voltage Power Devices?

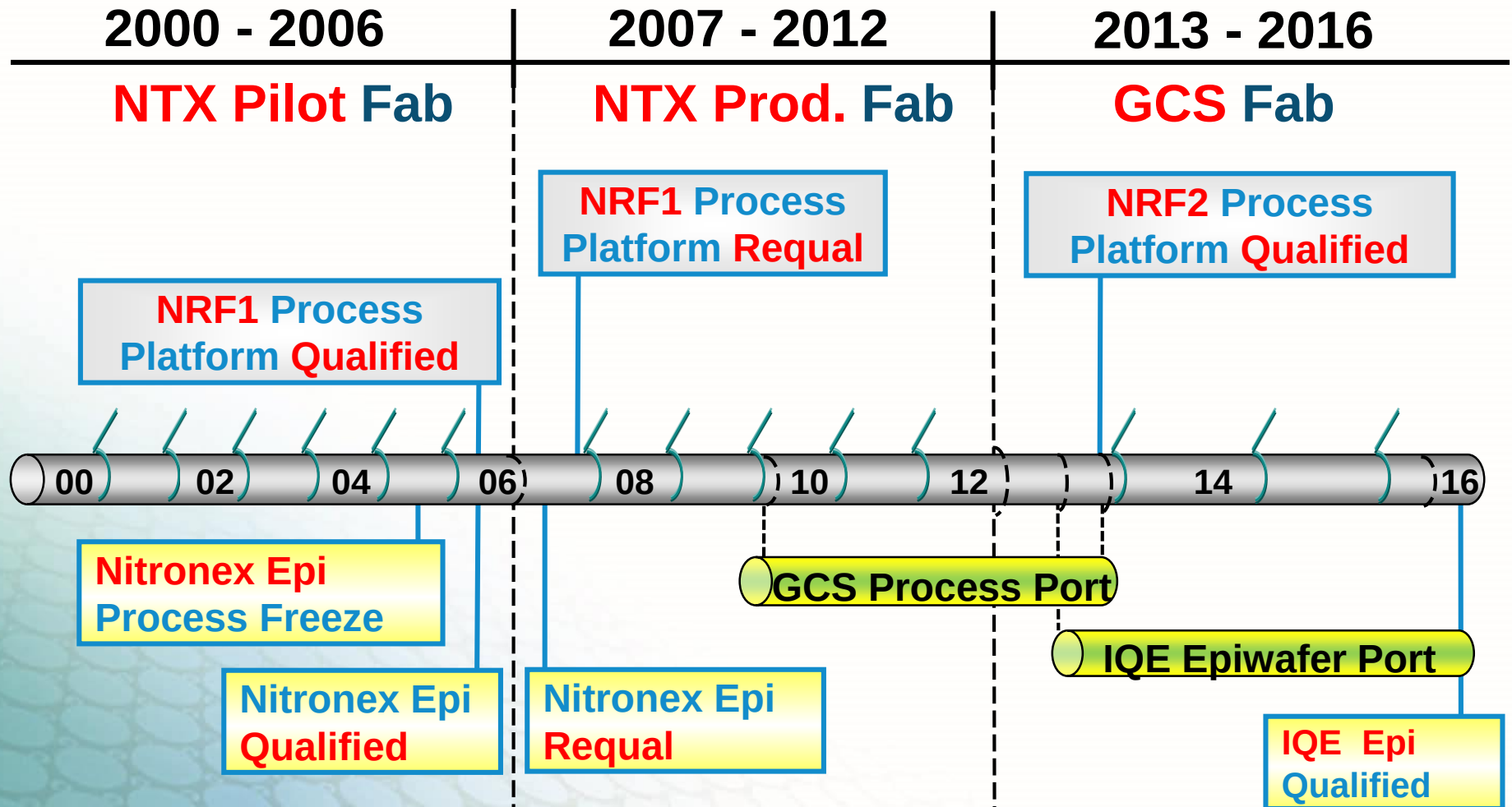


- **Similar** to the market pull for **GaN to replace Si based LDMOS** for commercial wireless applications, there is a similar phenomena occurring in the power device industry to **replace Si-based MOSFET and IGBT** with **GaN-on-Silicon HFET technology**.
- Adoption of GaN-on-Si Power Devices is 2-3 years ahead of GaN-on-Si RF devices for BTS applications.
- **GaN-on-SiC is not considered by the Power Device** market due to high costs. The market has already selected GaN-on-Si for 1200V and below.
- High-Volume: (**> 10k wafers/ wk**)
 - Wireless Infrastructure (**>1.5K 6-inch** wafers/wk by 2020)
 - RF Energy (**>7K 6-inch** wafers/wk by 2020)
 - Power Devices (**~19K 6-inch** wafers/wk by 2020)

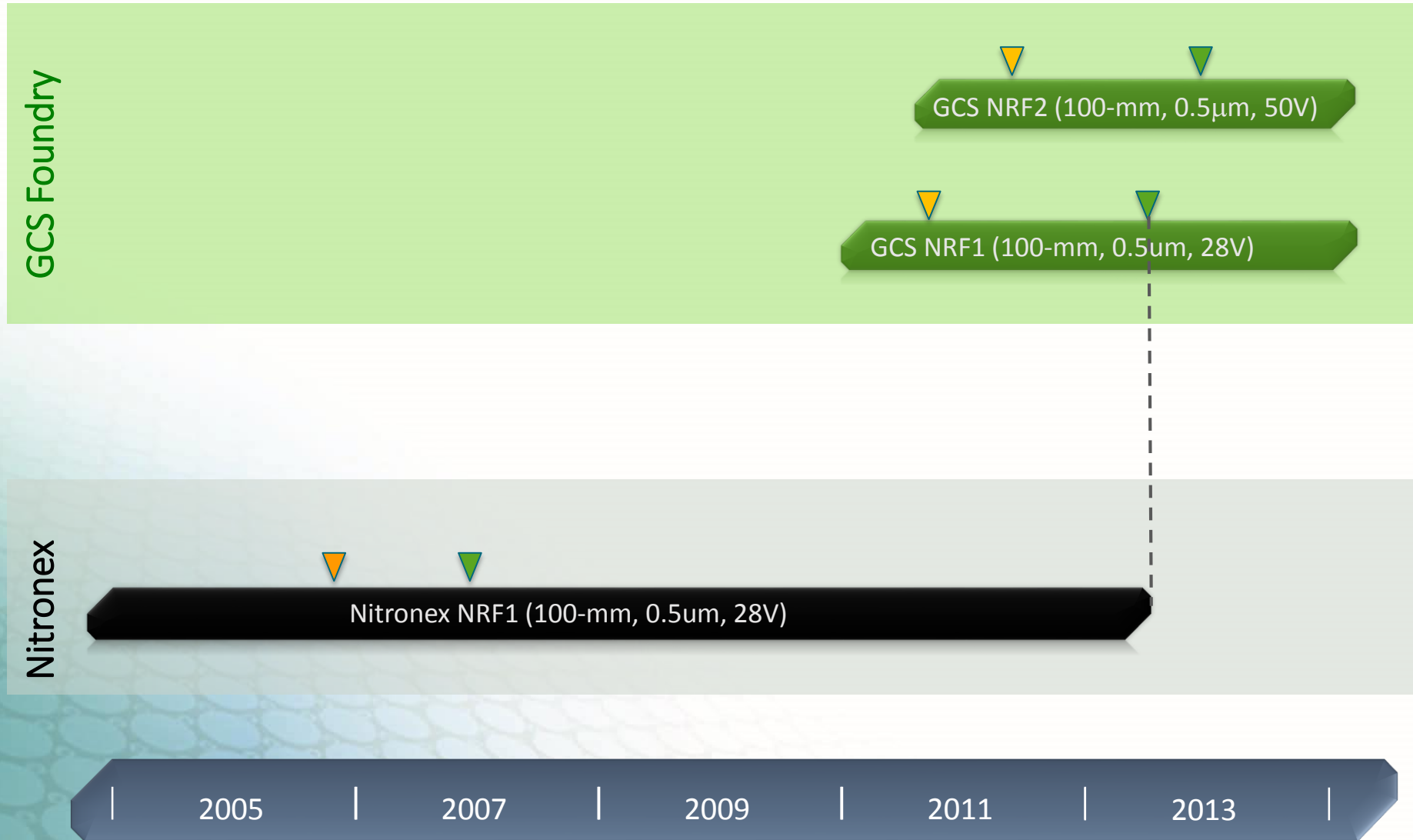
**> 27K
6-inch
wafers/week**

Could the SiC substrate supply chain really handle this volume of demand?

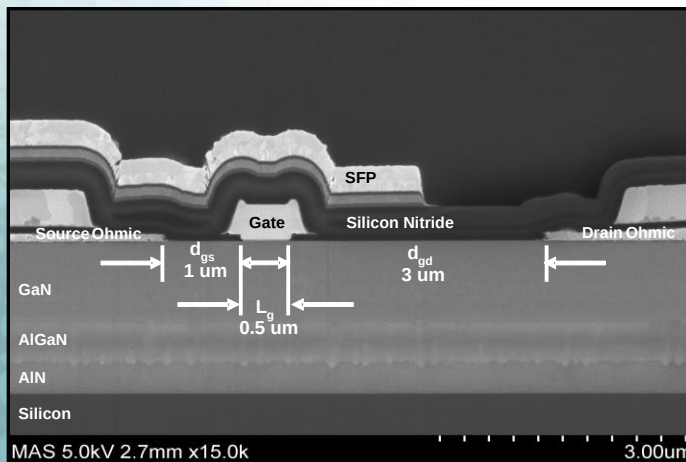
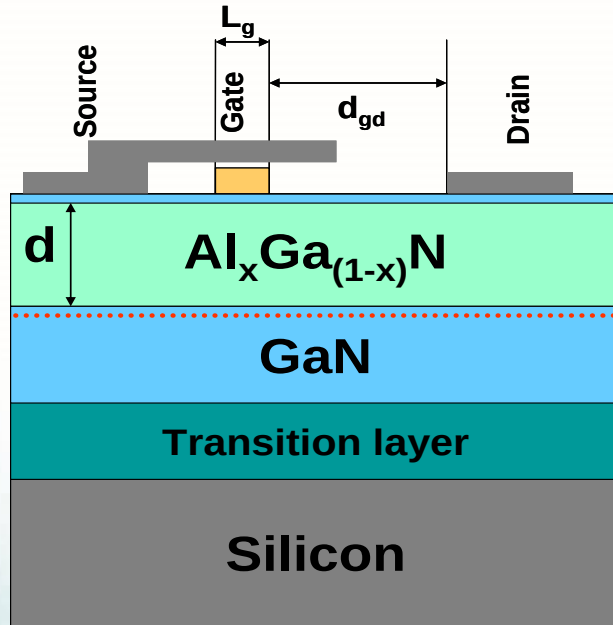
Nitronex GaN-on-Silicon Process Qualification



MTS GaN-on-Si Process *History*

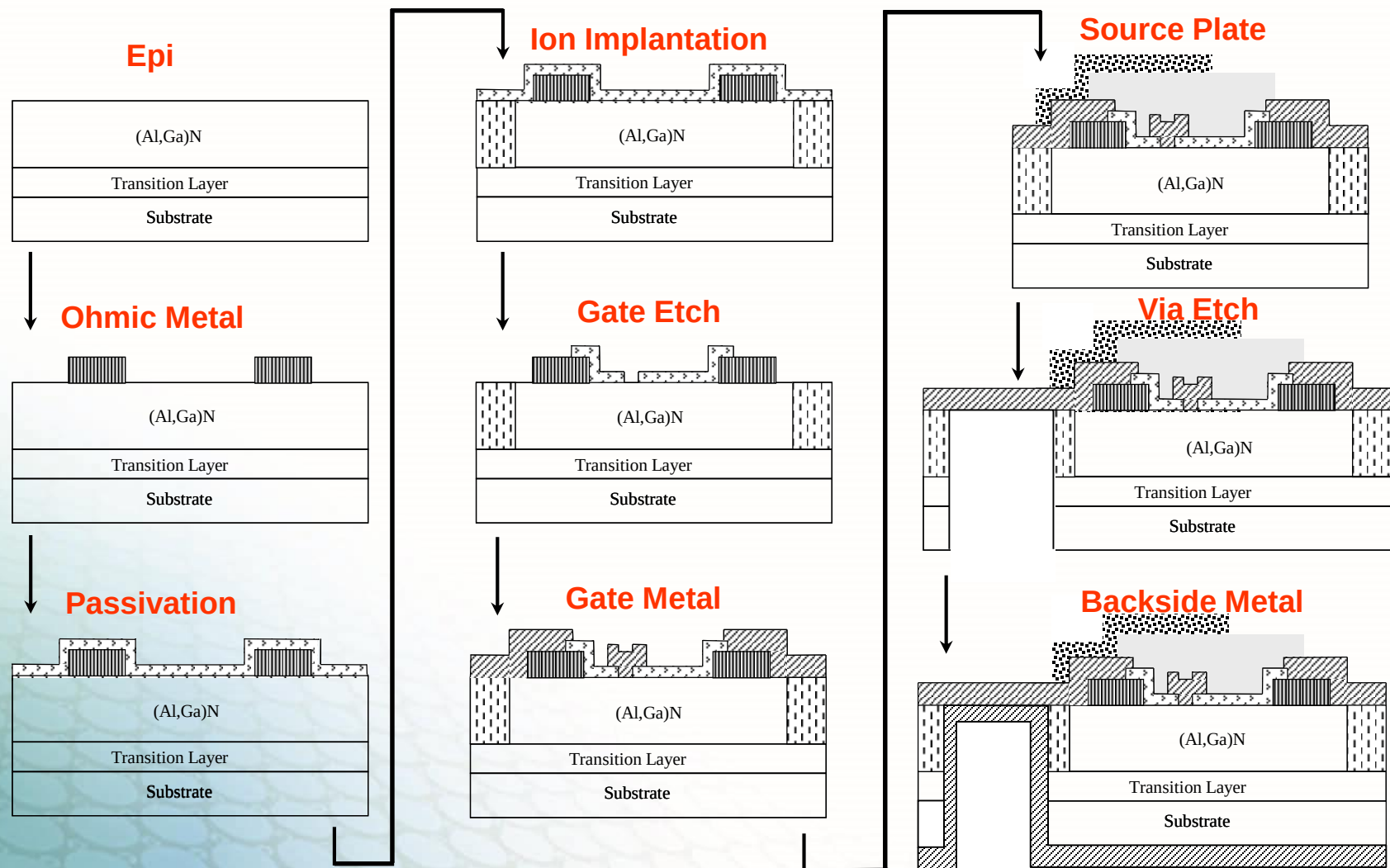


Nitronex 28V NRF1 Technology Attributes



- Nitronex Internal Supply Epiwafers
- Gate Length (L_g) = $0.5\mu\text{m}$
- Gate-Drain Spacing (d_{gd}) $\sim 3.2\mu\text{m}$
- Ni/Au Gate-Pitch = $30\mu\text{m}$
- Source Field Plate
- Air-bridge interconnect level
= $4\mu\text{m}$ electro-plated Au
- Parasitic Channel Area $> \sim 1\text{E}11$
- Silicon wafer thinned to $150\mu\text{m}$
- Backside via process $75\mu\text{m}$ diameter on source pads

NRF1 Device Fabrication Sequence



Similar to GaAs pHEMT process

NRF1 Technology Qualification 2006



NRF1 Technology Platform Qualification



Test Name	Test Standard	Sample Set	Result
3-Temp DC	JEP118-B	30 Devices/Temp @ 260, 285, and 310°C	Ea = 2.0eV, MTF > 10 ⁹ hours at 150°C
DC-HTOL	JESD22-A108	45 devices, 2000 hours	20-yr I _{max} drift <7% at 200°C 20-yr I _{max} drift <3% at 150°C
RF-HTOL	JESD22-A101-A	12 Devices, 500 hours	< 0.25dB drift through 500 hours
ESD-HBM	JESD22-A114	9	>1000V (class 1C) for NPT35050
ESD-MM	JESD22-A115	9	>200V (class M3) for NPT35050
Thermal Impedance	IR imaging	9	All samples meet datasheet target of 1.95°C/W for NPT35050
Autoclave	JESD22-A102	45	Minimal change in performance
VSWR	10:1 VSWR	5	Minimal change in performance
Temp. Cycling	JESD22-A104	45	Minimal change in performance
Thermal Shock	M-750-1056	15	Minimal change in performance
Solderability	JESD22-B102	4	Passed
Mech. Shock	M-883-2002	38	Passed
Vibration	M-883-2007	38	Passed
Const. Acceleration	M-883-2001	38	Passed
Moisture Res.	M-883-1004	38	Passed
Salt Atmosphere	M-883-1009	15	Passed
Solvent Res.	M-883-2015	15	Passed
Bond Strength	M-750-2037	15	Passed

• Qualification plan follows JEDEC and MIL standards

• List of tests based on standard LDMOS qualification reports

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General Product Qualification Plan

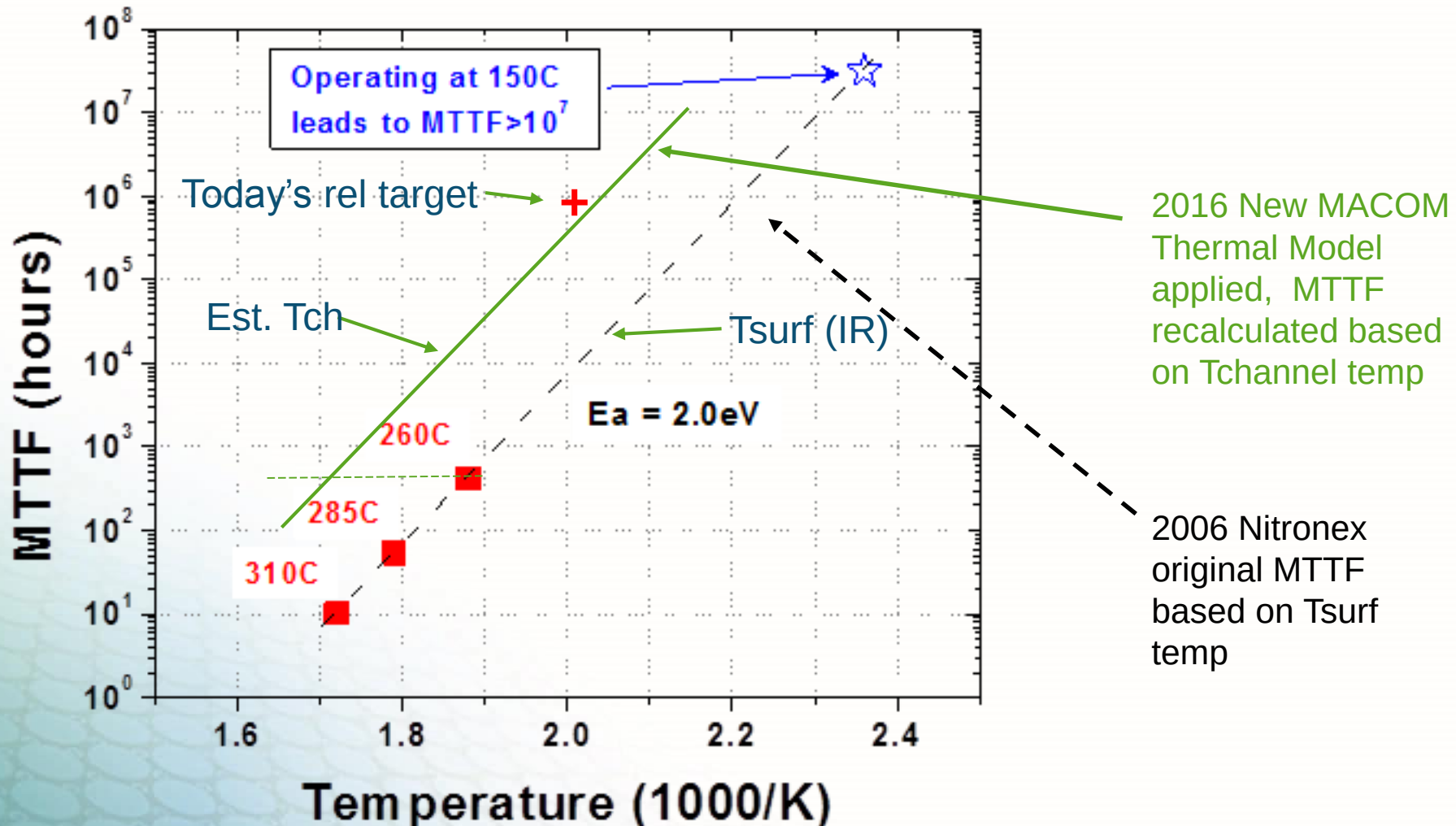


Stress	Abbr.	Reference	Stress Conditions	DUT Failure Criterion	Wafers	Units / wafer	Total units	Acceptance criterion
DC High temperature operating life	DC-HTOL	JESD22-A108, JESD85	VDS=28V, TJ=200C, 168hr	Catastrophic / hard failure or device parametric drift > 15% all DC parameters except 10X drift IDLK	3	16	48	0 fail
ESD Human Body Model	ESD-HBM	JESD22-A114	See reference	10X change in leakage or catastrophic fail	3	3	9	Classification
ESD Charged Device Model	ESD-CDM	JESD22-C101	See reference	10X change in leakage or catastrophic fail	3	3	9	Classification
ESD Machine Model	ESD-MM	JESD-A115	See reference	10X change in leakage or catastrophic fail	3	3	9	Classification
Temperature cycling	TC	JESD22-A104	-65/+150C	Datasheet specification limits	3	25	75	500 cycles / 0 fail
Mechanical shock + vibration variable frequency + constant acceleration	MS+VVF+CA	M-883-2002 cond. B + -2007 cond. A + -2001 cond. B	See reference	Catastrophic / hard failure or device parametric drift > 15% all DC parameters except 10X drift IDLK	1	15	15	0 fail
Moisture resistance	MR	M-883-1004	See reference	Visual inspection per MIL spec only (mechanical samples)	1	15	15	0 fail
Salt atmosphere	SA	M-883-1009	Condition "A" - see reference	Visual inspection per MIL spec only (mechanical samples)	1	15	15	0 fail
TOTAL							195	

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- Note: NRF1 & NRF2 customer requirements (primarily military) do not require HAST/THB specification nor HTRB as part of qualification

2006 NRF1 Nitronex ALT Results



2016 New MACOM Thermal Model applied, MTTF recalculated based on Tchannel temp

2006 Nitronex original MTTF based on Tsurf temp

- NRF1 Process met 2006 reliability goal of $>1\text{E}7$ hours at $T_{\text{surf}}=150^\circ\text{C}$.
- NRF1 does not meet today's reliability goal of $1\text{E}6$ hours at $T_{\text{ch}}=225^\circ\text{C}$.

GaN-on-Si Process History



NRF1 Technology Qualification

- NPT35050 was selected as the Test Vehicle for the NRF1 Technology Qualification

NPT35050A Qualification Document

Revision 2

Production Qualified on 10/10/2006!

Nitronex

Nitronex NRF1 (100-mm, 0.5um, 28V)

Product Selection Guide

2.5 GHz RF Power Transistors									
Part Number	Package	Band	Supply Voltage (V)	CW P _{out} @ 28dB (W)	CW Drain Efficiency @ P _{out} (%)	CW Power Gain (dB)	Linear Output Power 802.15e (W)	Drain Efficiency 802.15e (%)	Notes
NPT25050	P0150S	2.3-2.7	28	25	58	14	1.5	28	
NPT25100	AC780B	2.3-2.7	28	80	62	16	10	24	
NPT25150	AC780B	2.3-2.7	28	100	62	16	10	24	
NPT25200	AC780B	2.3-2.7	28	120	62	16	10	24	
(1) 2.0W typical EVM									
3.5 GHz RF Power Transistors									
Part Number	Package	Band	Supply Voltage (V)	CW P _{out} @ 28dB (W)	CW Drain Efficiency @ P _{out} (%)	CW Power Gain (dB)	Linear Output Power 802.15e (W)	Drain Efficiency 802.15e (%)	Notes
NPT35050	P0150S	3.3-3.8	28	18	45	10.5	1.5	18	
NPT35100	AC200B	3.3-3.8	28	60	50	12	6.3	18	
Broadband Power Transistors									
Part Number	Package	Frequency Band (GHz)	Supply Voltage (V)	Output Power Part @ 30dB (W)	PAE (dB)	Small Signal Gain @ 30dB (dB)	Drain Efficiency (%)	Notes	
NPTB0500	P0150S	DC-6.0	28	5	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	20	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	40	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	60	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	80	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	100	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	120	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	140	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	160	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	180	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	200	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	220	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	240	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	260	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	280	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	300	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	320	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	340	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	360	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	380	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	400	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	420	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	440	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	460	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	480	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	500	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	520	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	540	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	560	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	580	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	600	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	620	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	640	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	660	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	680	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	700	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	720	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	740	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	760	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	780	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	800	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	820	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	840	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	860	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	880	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	900	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	920	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	940	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	960	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	980	25	10	15	85	
NPTB0500	AC200S	DC-6.0	28	1000	25	10	15	85	

2005

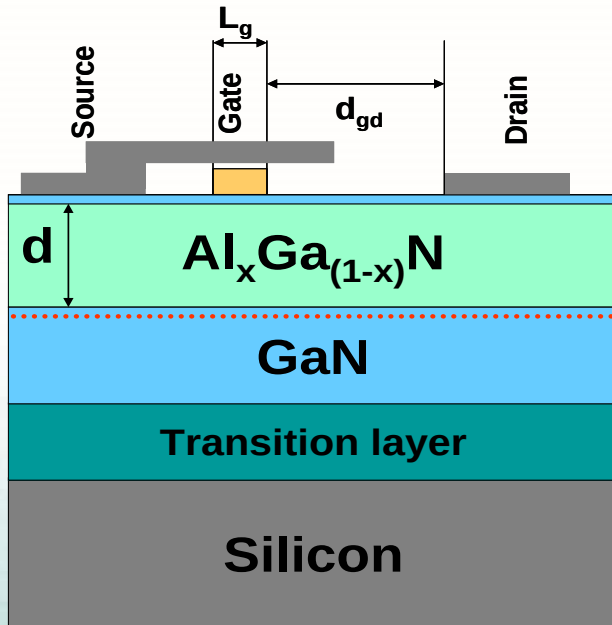
2007

2009

2011

2013

Nitronex 50V NRF2 Technology Attributes

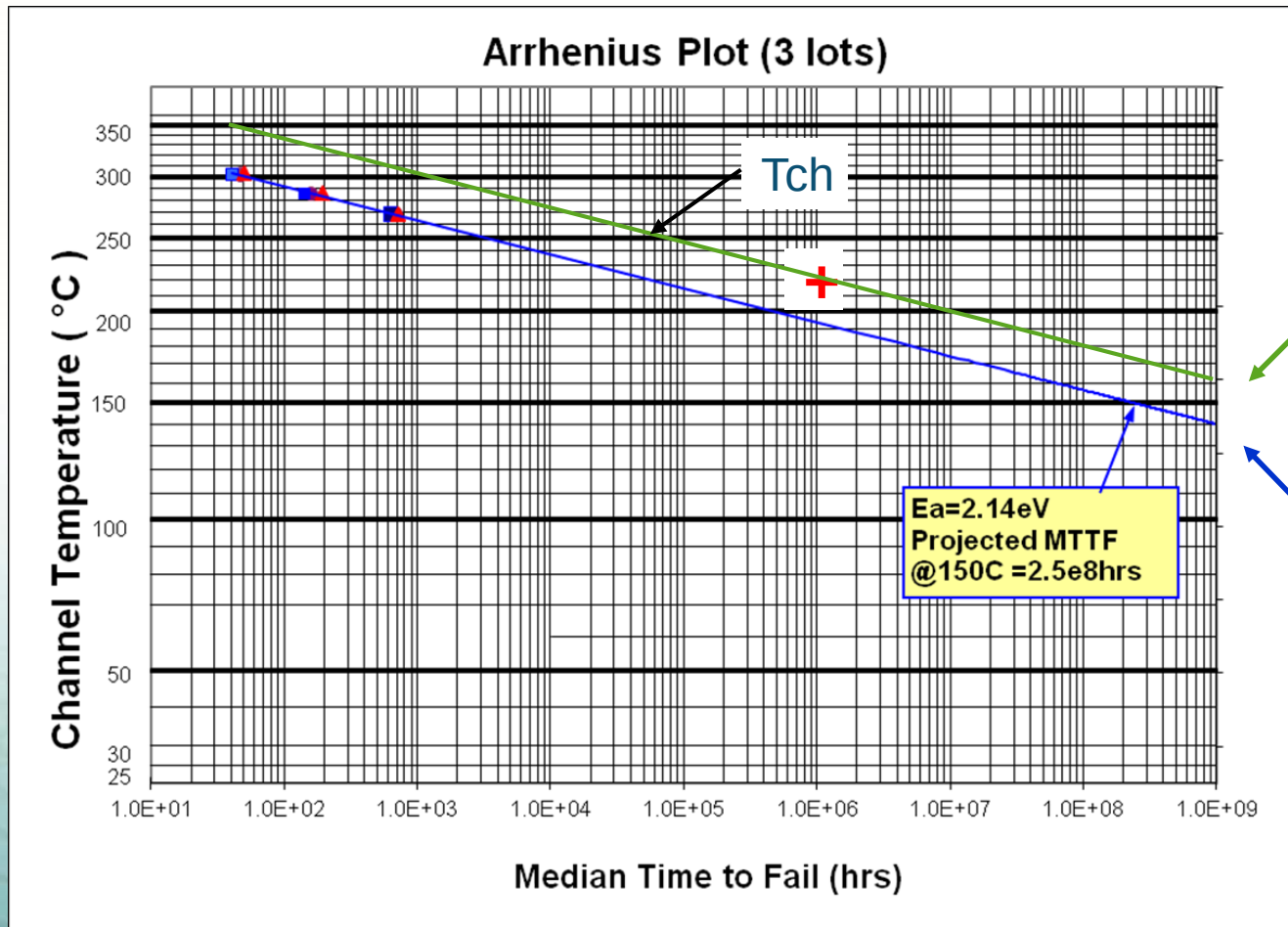


Continuous Improvement

- Process sequence and conditions
- Thinner die (thermal)
- 50V design rules

- Nitronex Internal Supply Epiwafers
- Gate Length (L_g) = $0.5\mu\text{m}$
- Gate-Drain Spacing (d_{gd}) ~ $4.6\mu\text{m}$
- Ni/**Pd**/Au Gate-Pitch = $50\mu\text{m}$
- Source Field Plate
- Air-bridge interconnect level
= $4\mu\text{m}$ electro-plated Au
- Parasitic Channel Area > $\sim 1\text{E}11$
- Silicon wafer thinned to $50\mu\text{m}$
- Backside via process $75\mu\text{m}$ diameter on source pads

2013 NRF2 GCS ALT Results



2016 New MACOM Thermal Model applied, MTTF recalculated based on Tchannel temp

2013 Nitronex original MTTF based on Tsurf temp

- Met 2013 reliability goal of $>1\text{E}8$ hours at $T_{\text{surf}}=150\text{C}$.
- Meets the current reliability goal of $1\text{E}6$ hours at estimated $T_{\text{ch}}=225\text{C}$.

GaN-on-Si Process History



GCS Foundry

Nitronex

DOC ID:	NGO-153	TITLE: NRF2 TECHNOLOGY QUALIFICATION DOCUMENT	PAGE:	1 OF 8
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NRF2 Technology Qualification Report

Revision 1

Production Qualified 2013

NITRONEX LLC

INTERNAL REVIEW

THIS COPY MAY NOT REPRESENT THE CURRENT REVISION OF THIS DOCUMENT.
VALID CURRENT REVISIONS OF DOCUMENTS CAN BE FOUND IN THE NITRONEX MASTER/CONTROL DOCUMENT CONTROL SYSTEM.

GCS NRF2 (100-mm, 0.5 μ m, 50V)

GCS NRF1 (100-mm, 0.5 μ m, 28V)

n, 28V)

2005

2007

2009

2011

2013

1st Large Rate Volume Production



Nitronex NRF1 Technology in High Volume Production



Harris Corporation Demonstrates JTRS Battlefield Networking Technology on Falcon III Manpack Radios

JTRS-approved AN/PRC-117G Radio Transmits Voice over IP (VoIP), Live Video and Text over Harris Network Using Key Government Wideband Waveforms

WASHINGTON, D.C., October 5, 2009 (AUSA Booth #1439) — Harris Corporation (NYSE:HRS), an international communications and information technology company, has demonstrated the exchange of high-bandwidth voice and data over a tactical network using wideband waveforms developed by the Joint Tactical Radio System (JTRS) program. The demonstration was held at the U.S. Army (AUSA) Meeting and Exposition Oct. 5-7 in Washington, D.C.

"Our demonstration illustrates the capability of the AN/PRC-117G radio to form a network of radios that can connect air, ground and satellite assets. This radio is compliant with the JTRS Software Defined Radio (SDR) standards approach to radio technology," said Steve Marschilok, President, Defense Systems, Harris Corporation. "Through these networks, which will operate advanced wideband and legacy narrowband waveforms, we can connect our deployed personnel in numerous ways so they can take advantage of critical information. The demonstrations involved a network of AN/PRC-117G radios operating the JTRS Wideband Networking Waveform (WNW) and the JTRS Wideband Networking Waveform (WNW) and the JTRS Wideband Networking Waveform (WNW). Harris used the three waveforms to bear Voice over Internet Protocol (VoIP), live video, tactical chat, and tactical ground reporting (TIGR) traffic."

The demonstration was made possible through the JTRS Enterprise Business Model. Harris utilized access to current versions of the WNW and SRW waveforms from the JTRS Information Repository to develop the demonstration.

Customer-Focused Technology: Falcon III AN/PRC-117G

The Falcon III AN/PRC-117G is the first JTRS-approved tactical radio to offer reliable and secure access to high-bandwidth applications in addition to narrowband legacy waveforms. The radio is certified by the Joint Interoperability Test Command (JITC) as compliant with Demand Assigned Multiple Access (DAMA) military standards and upgradeable to access the Mobile User Objective System (MUOS), both for transmitting high-bandwidth voice and data over tactical military satellites.

The AN/PRC-117G also includes a Remote Operated Video Enhanced Receiver (ROVER) interoperable mode that provides Air Force personnel on the battlefield with the ability to receive live video directly from unmanned aerial vehicles (UAVs). The AN/PRC-117G previously received Type-1 information security certification from the National Security Agency.

Harris RF Communications is the leading global supplier of secure radio communications and embedded high-grade encryption solutions for military, government and commercial organizations. The company's Falcon family of software-defined tactical radio systems encompasses manpack, handheld and vehicular applications. Falcon III is the next generation of radios supporting the U.S. military's Joint Tactical Radio System (JTRS) requirements, as well as network-centric operations worldwide. Harris RF Communications is also a leading supplier of assured communications systems and equipment for public safety, utility and transportation markets - with products ranging from the most advanced IP voice and data networks to portable and mobile single- and multi-band radios.



Powered by Nitronex NPTB00004 and NPTB00025

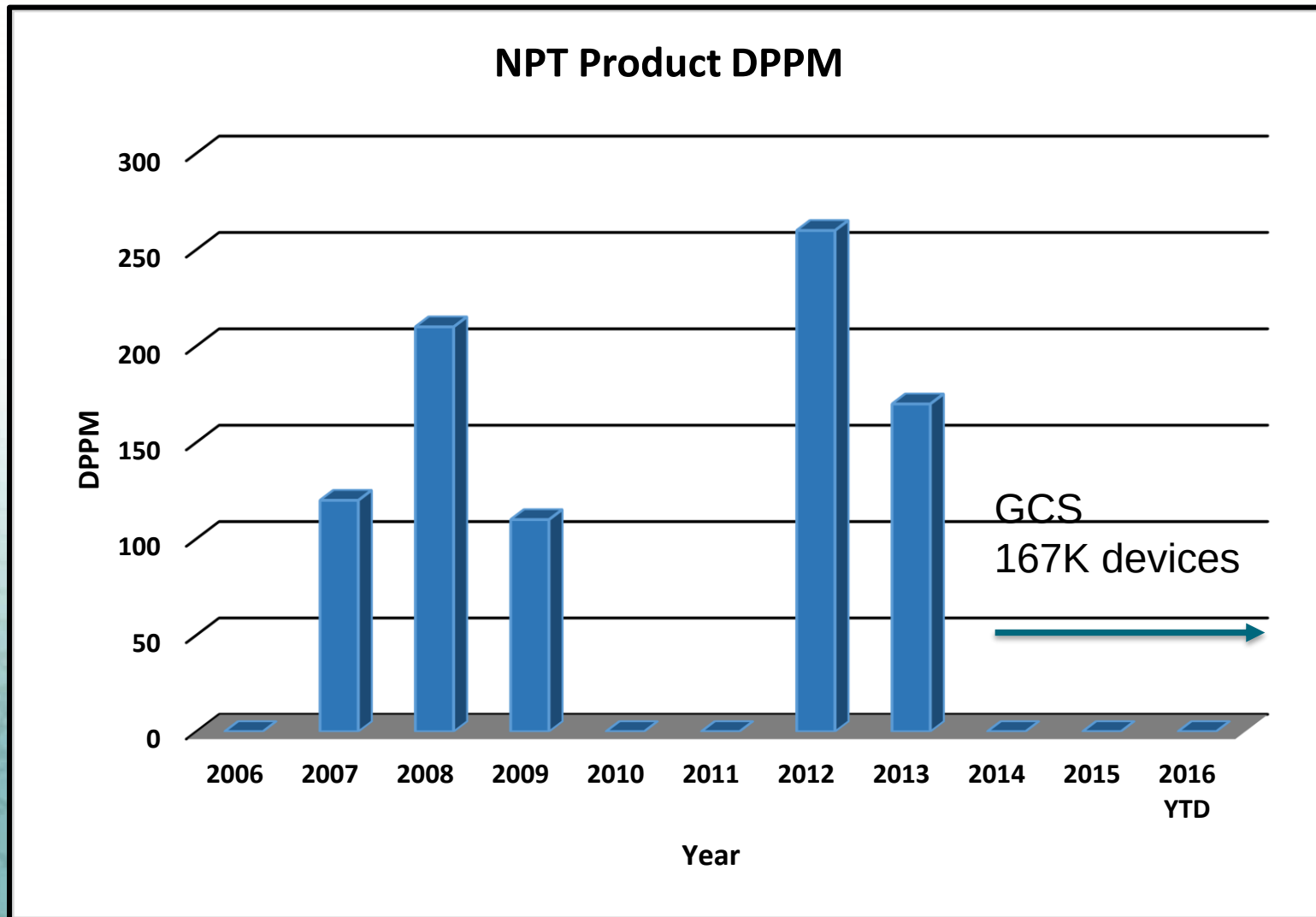
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Between 2006 and 2016, over 1M devices have been delivered using NRF1 & NRF2

GaN/Si DPPM History – updated May 2016



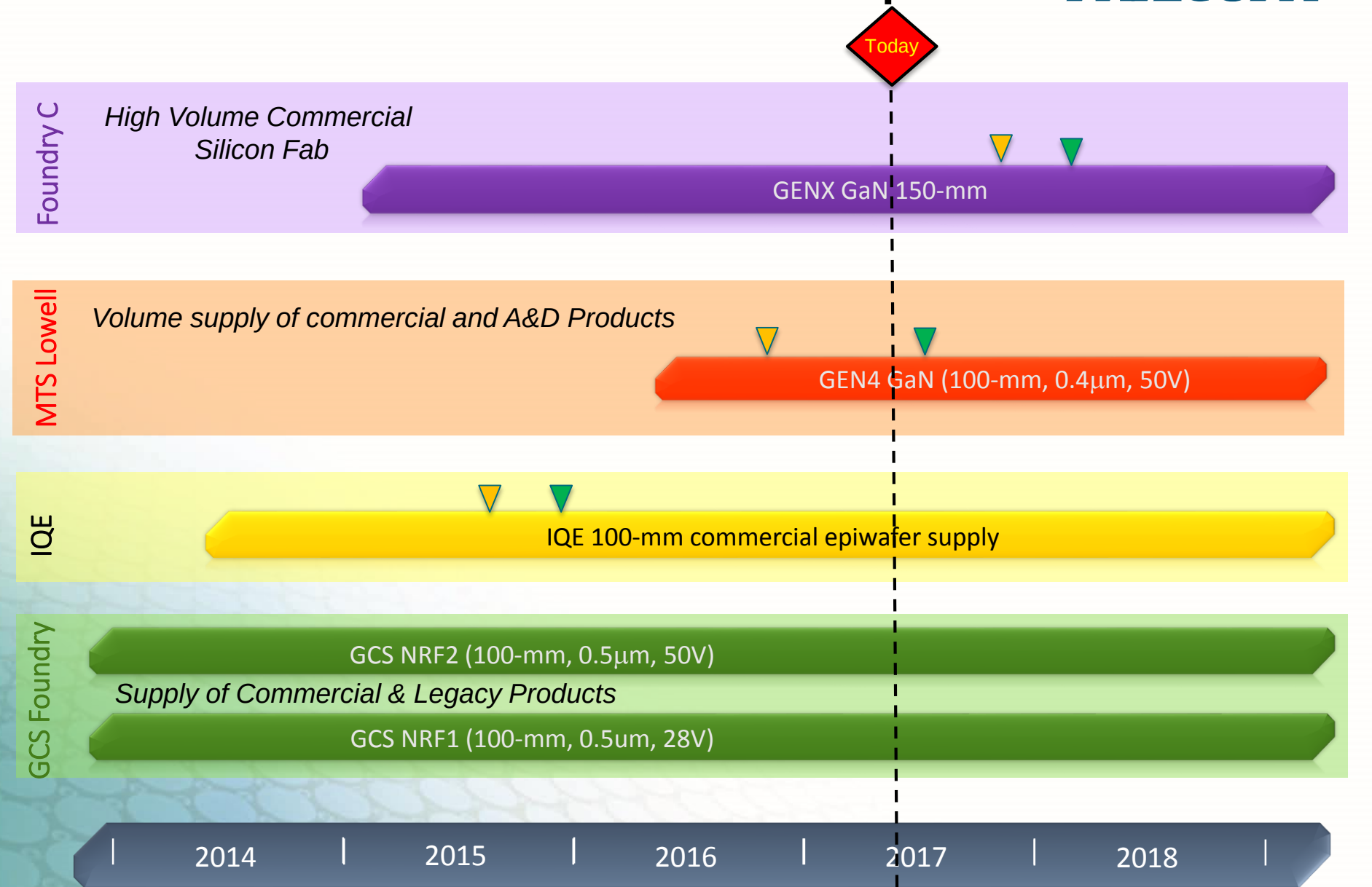
1.01M Devices Total





- In 2014, MACOM Purchases Nitronex from GaAs Labs
- Technology Development shifts away from GaN Low Volume Military Application Strategy and focuses on High Volume Commercial Wireless & RF Energy
- Technology Development Team starts works on Gen4 and qualification of the MACOM Lowell fab.

MTS GaN-on-Si *Current* Process Development



IQE Chosen as Key Supply Chain Partner for MACOM's Global Gallium Nitride on Silicon Program



01 April 2014

Cardiff, UK, 01 April 2014: IQE plc (AIM: IQE; "IQE", "the Group") has entered an agreement with M/A-COM Technology Solutions Inc. ("MACOM"), a leading supplier of high performance RF, microwave, and millimeter wave products, to deliver Gallium Nitride on Silicon (GaN-on-Si) performance with 200mm silicon cost structure.

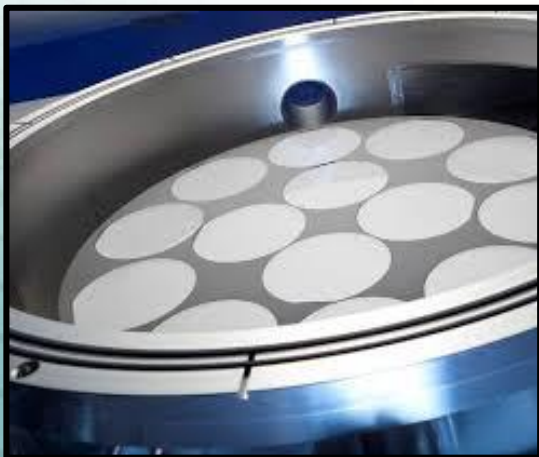
The agreement was announced last night by MACOM along with the introduction of an IP licensing program that will enable GaN insertion and large-scale production across the RF industry. As part of the licensing program, MACOM will make available its GaN-on-Si technology to select companies for use in RF applications.

In 2014, MACOM outsourced GaN-on-Si epiwafer manufacturing to a high volume Epi supplier

High Volume Epiwafer Supplier Qualified

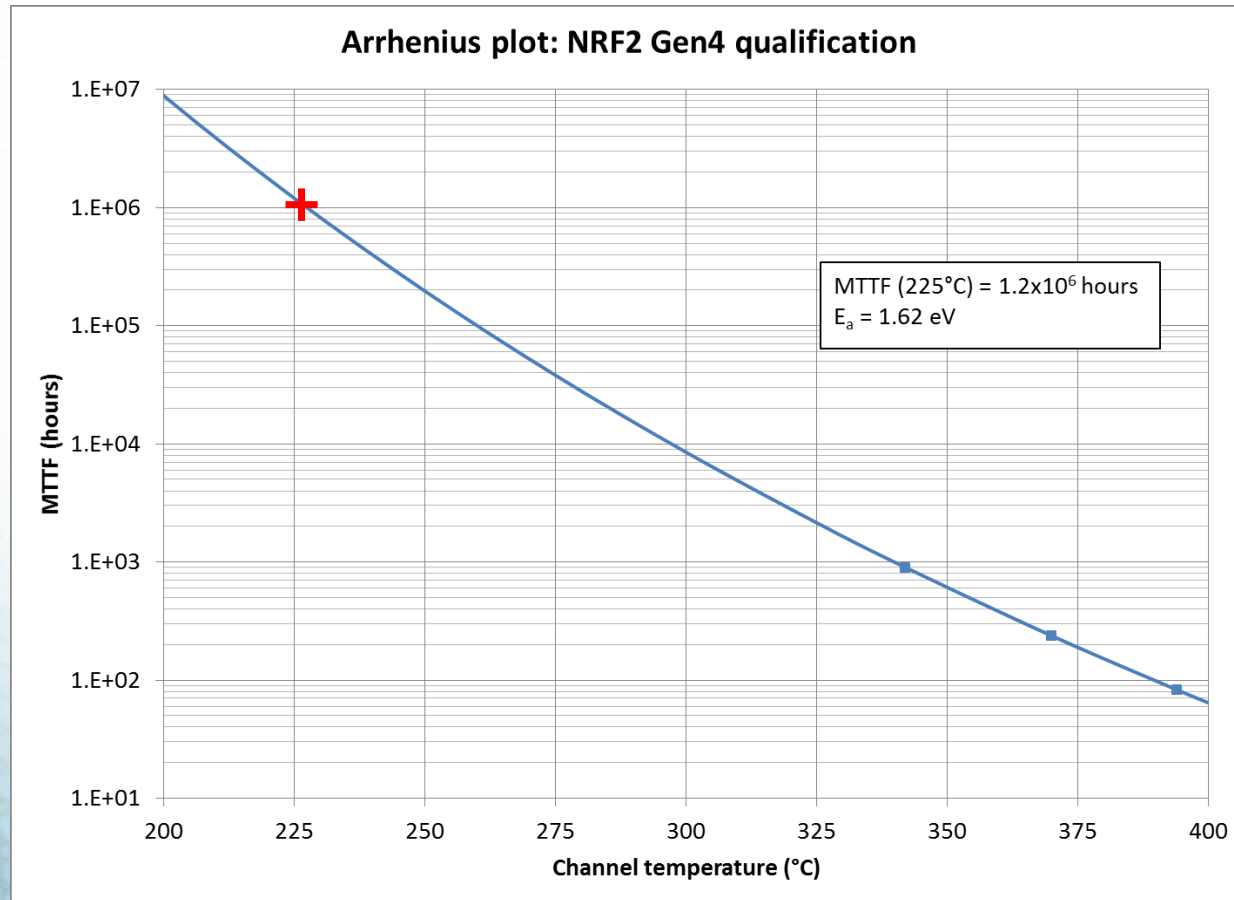


- Same commercial tooling supporting GaN based LED production and Power Switching applications
- Multiple 100-mm, 150-mm and 200-mm growth capability
- Redundant sources
- Exclusive license to only supply MACOM for GaN-on-Si RF power
- VP of IQE's GaN growth business unit is an ex-Nitronex employee



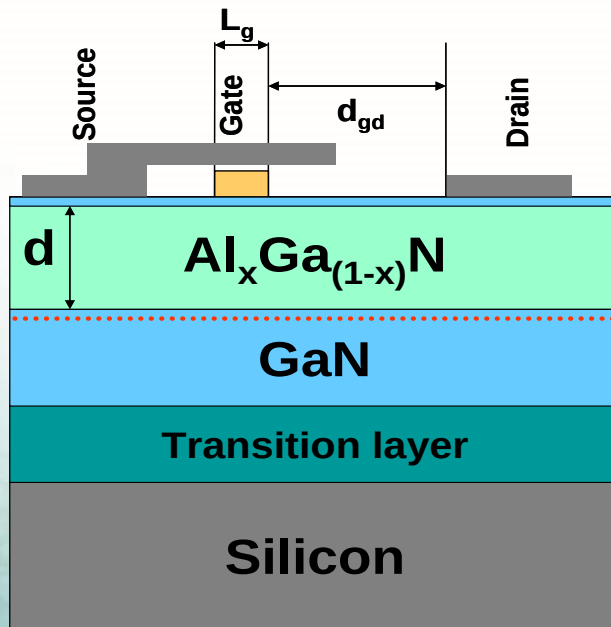
2015 IQE Epi Qualification ALT Results **MACOM**

- Used GCS NRF2 Foundry Process



- Meets reliability goal of 1E6 hours at Tch=225C

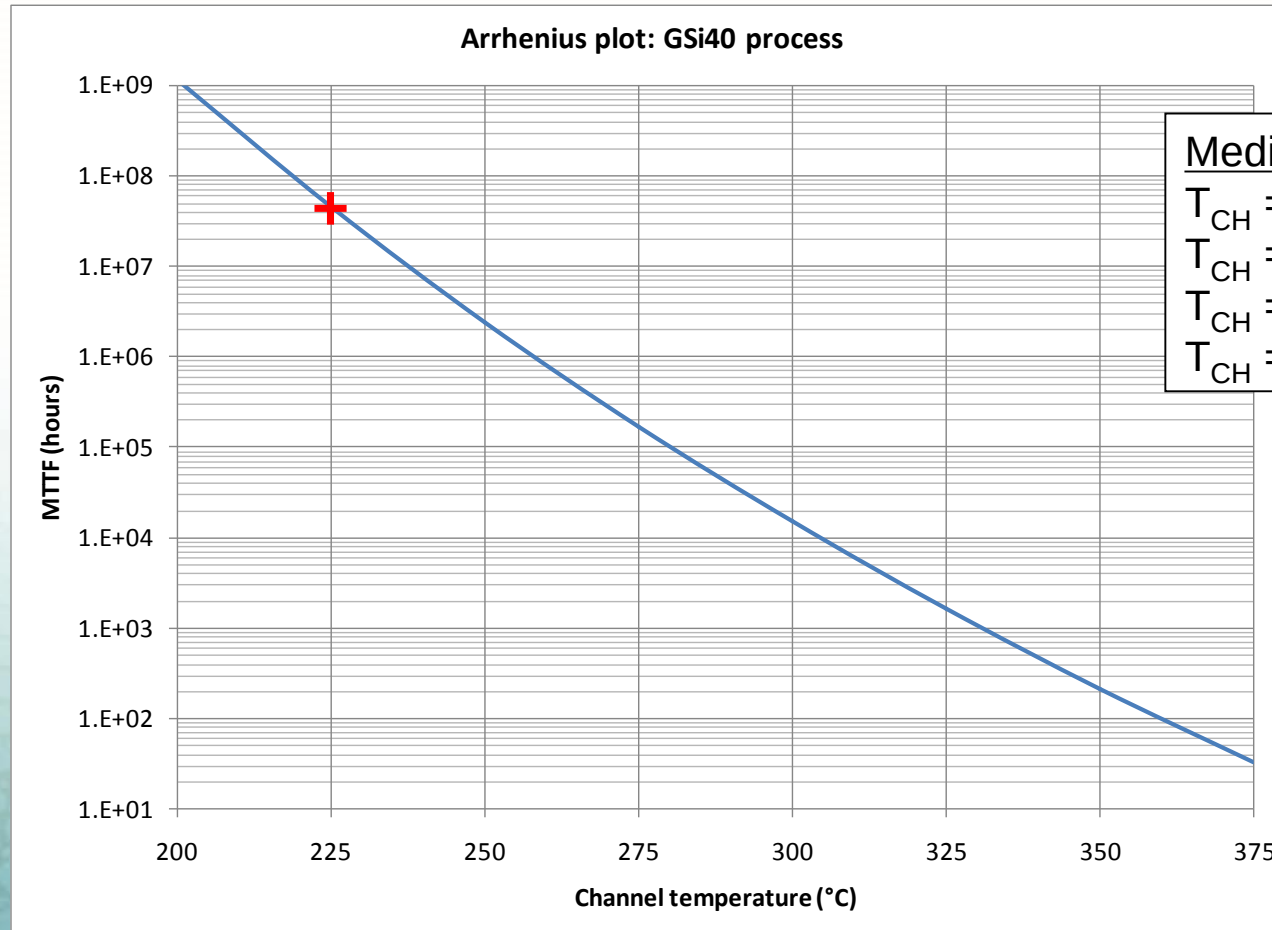
MACOM Gen4 Technology Attributes



Attribute	NRF1 (2006)	NRF2 (2013)	Lowell Gen4 (2017)
Epiwafer Supply	Nitronex	Nitronex	IQE
Gate Length	0.5um	0.5um	0.4um
Gate-Drain Spacing	3.2um	4.6um	4.6um
Gate Metal & Pitch	Ni/Au 30um	Ni/Pd/Au 50um	Ni/Pd/Au Various
Interconnect	Air bridge plated Au 4um	Air bridge plated Au 4um	Air bridge evaporated Au 3um
Parasitic Channel Area	~1E11	~1E11	<1E10
Wafer Thickness	150um	50um	50um
Backside Via	75um pad via	75um pad via	30x60um ISV
Foundry	Nitronex/GCS	GCS	GCS/MACOM
MTTF @225C	5E5 hrs	1E6 hrs	~4.5E7 hrs*

* Gen4 3-Temp ALT still in progress, see next page

2017 MACOM Lowell Gen4 ALT Results-to-date (Testing not complete)



Median failure times so far:

$T_{CH} = 308^{\circ}\text{C}$: 3,537 hours

$T_{CH} = 326^{\circ}\text{C}$: 1,301 hours

$T_{CH} = 343^{\circ}\text{C}$: 423 hours

$T_{CH} = 355^{\circ}\text{C}$: 70 hours

Test is ongoing due
to waiting for
required failures of
20% degradation

- Based on current status: Exceeds reliability goal of 1E6 hours at $T_{ch}=225^{\circ}\text{C}$
- Current Estimate: $\text{MTTF}(225^{\circ}\text{C}) = 4.5 \times 10^7$ hours; $E_a = 2.6$ eV

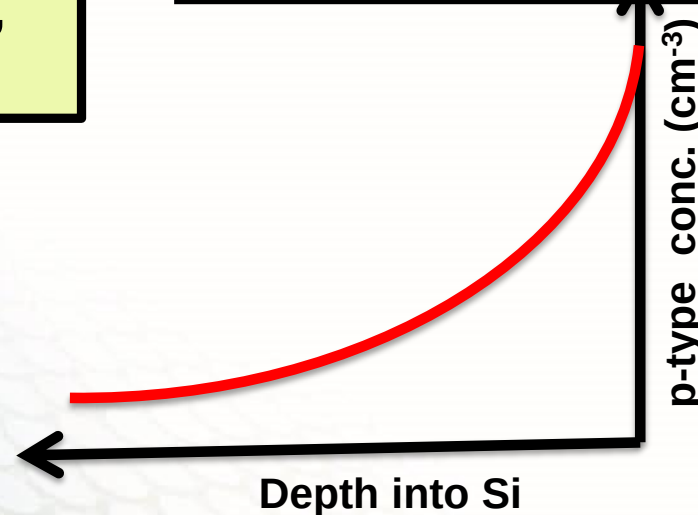
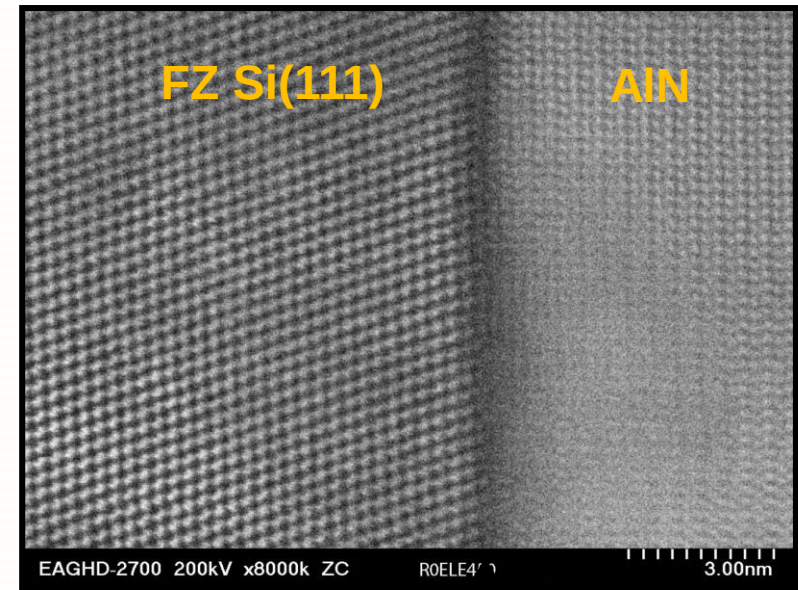
Parasitic Channel

What is it?

A region of p-type conductivity extending from Si substrate surface down into the Si.

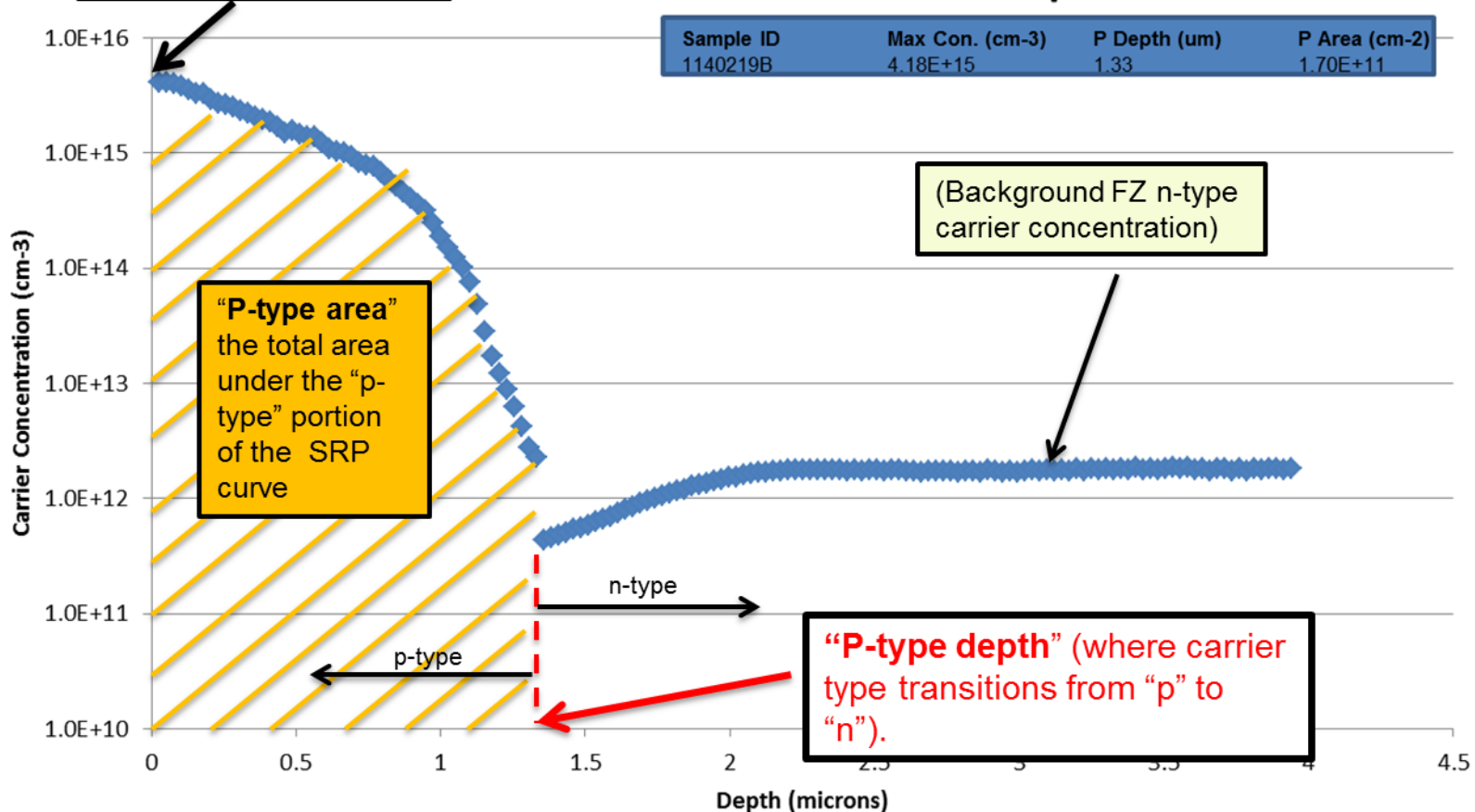
What does it do?

Couples to RF fields and presents a parasitic load during RF operation. Can decrease power, efficiency and gain.



Parasitic Channel Area Definition

Carrier Concentration vs Depth

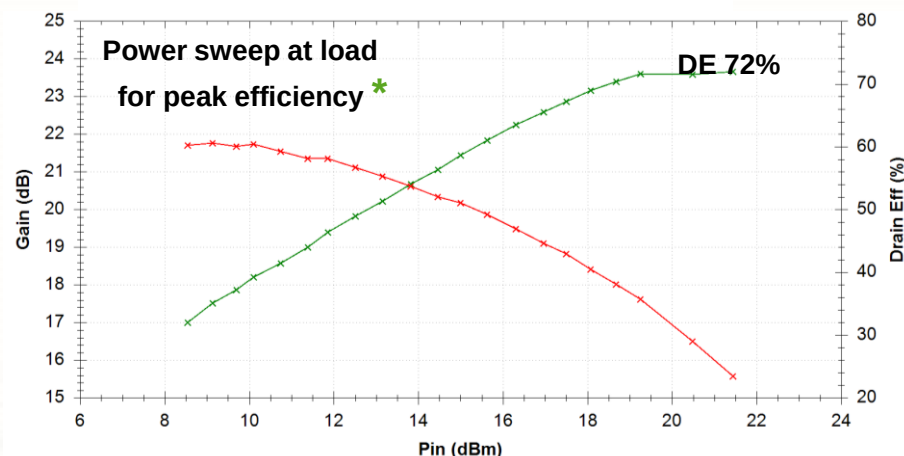
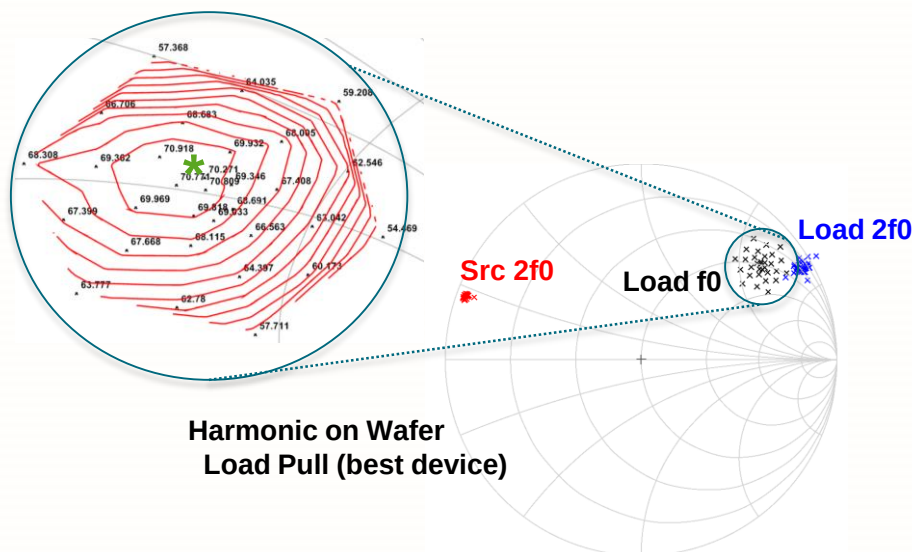
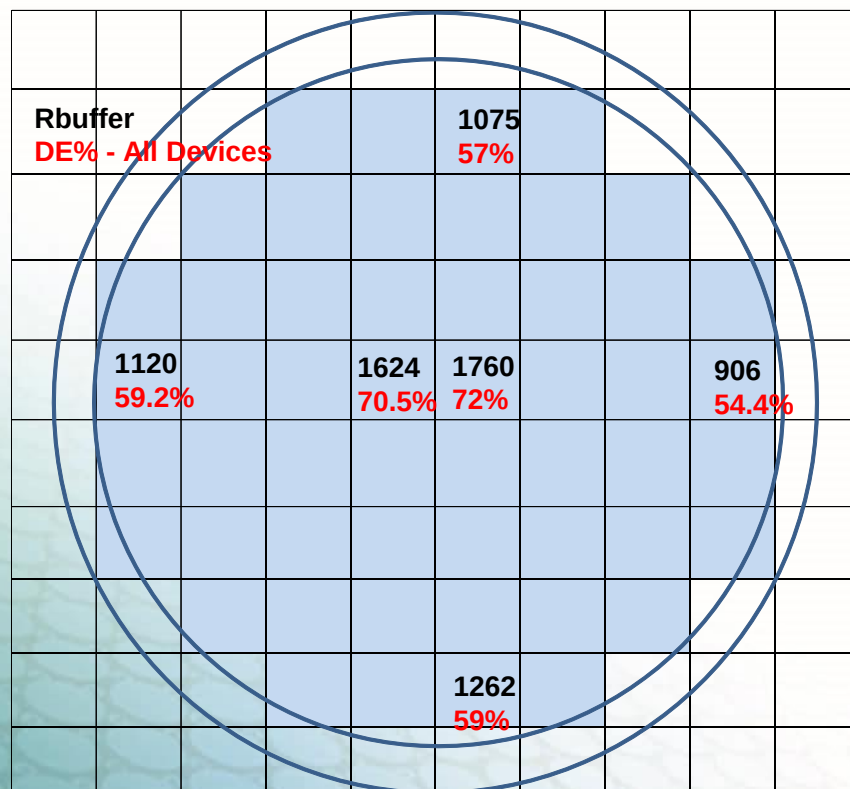


Correlation to Drain Efficiency



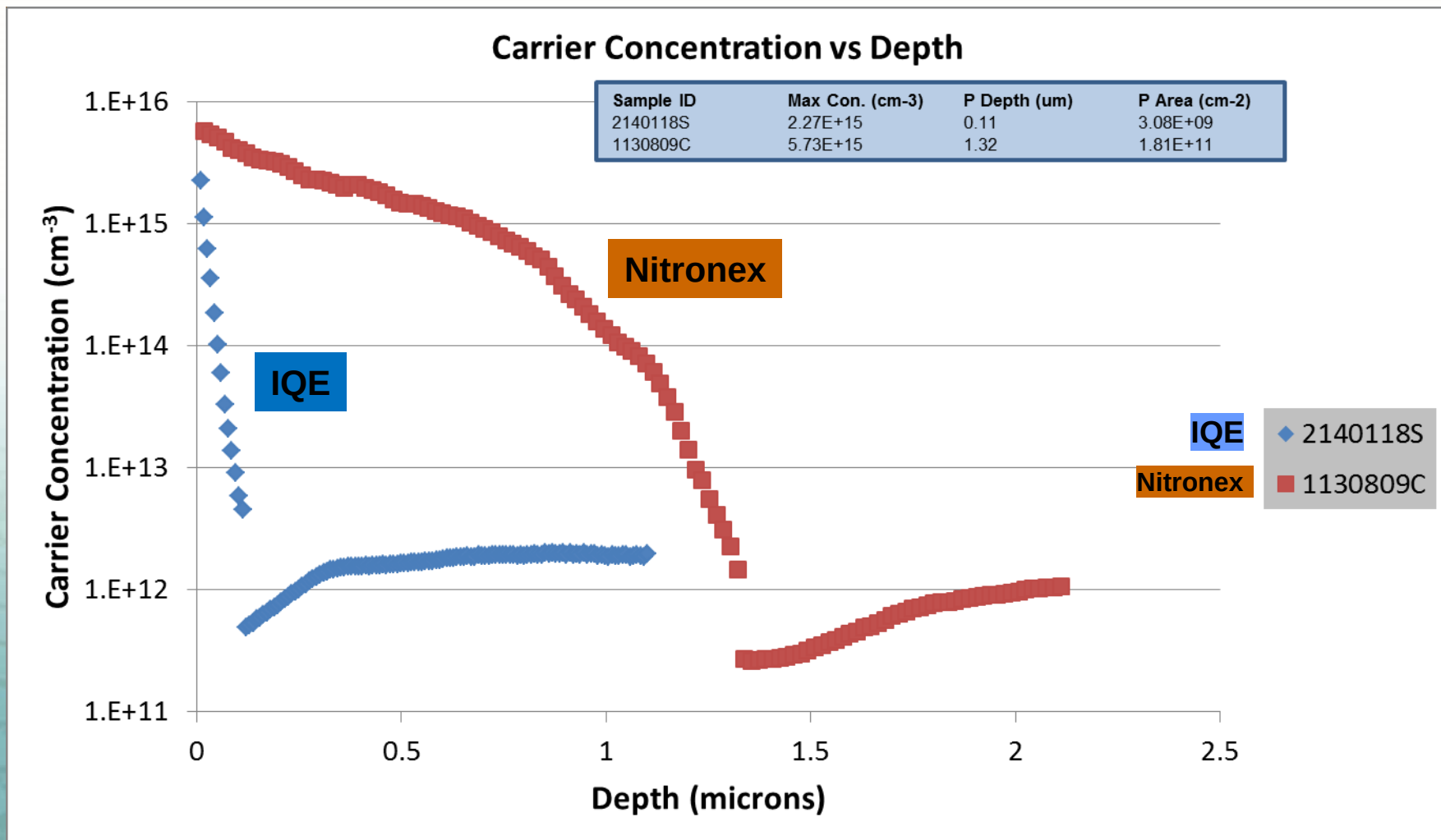
1st GaNSi Wafer (WN024E002-05)

Extraction of Parasitic Channel



Drain Efficiency correlated to Rbuffer/Parasitic Channel Area

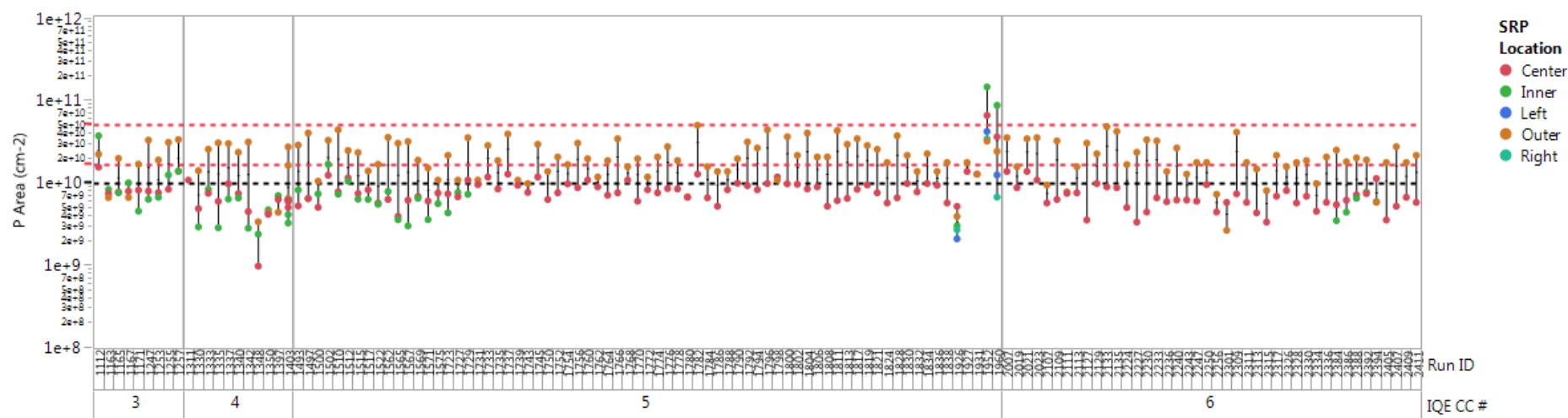
Nitronex Legacy Epi (NRF1 & NRF2) vs. IQE Epi vs. Gen4



IQE Parasitic Channel “Area” (cm⁻²)



IQE GaN-on-Silicon Epiwafer Parasitic Channel Trend Chart



- One wafer from each epi growth run is destructively tested for parasitic channel using Spreading Resistance Profiling.
- Chart represented above shows trend over 4 IQE growth campaigns.

Transistor Technology Roadmap

5G PA	3.5 GHz	3.95 GHz	4.65 GHz	5.8 GHz	28GHz Europe	28GHz Global	37GHz-40Ghz
Generation (name or number) and technology, foundry	MACOM GaN Si Gen 4 MACOM	MACOM GaN Si Gen 4B MACOM	MACOM GaN Si Gen 4B MACOM	MACOM GaN Si Gen 5 MACOM	MACOM GaN Si Gen L3 MACOM	MACOM GaN Si Gen L3 MACOM	MACOM GaN Si Gen L3 MACOM
Development status [In road map, Planned, In Development, Released]	In Qualification	In Characterization	In Characterization	In Development	In Development	In Development	In Development
Release date/year (Process qualified)	Q3 2016	Q3 2017	Q3 2017	Q3 2018	Q4 2018	Q4 2018	Q3 2019
Peak power [P3dB] CW sweep:25ms sweep time, period 200ms /die mm	6W/mm	6W/mm	6W/mm	6W/mm	4W/mm	4W/mm	3W/mm
Max stable Gain [dB] (small-signal)	23	23	22	20	13	13	12
Efficiency @ P3dB [supply voltage max]	75	75	70	70	45	45	40
Cutoff frequency Ft [GHz] (@ Vdd max)	14	16	16	18	40	40	50

Agenda

- ✓ GaN update in general
 - **Status of CMOS fab transfer**
 - Epi update
 - Process Corner lots
 - Reliability update – bHAST+
 - Idq Drift
 - Epiwafer technology development lab tour

High Volume CMOS FAB Partner



- **Motivation**

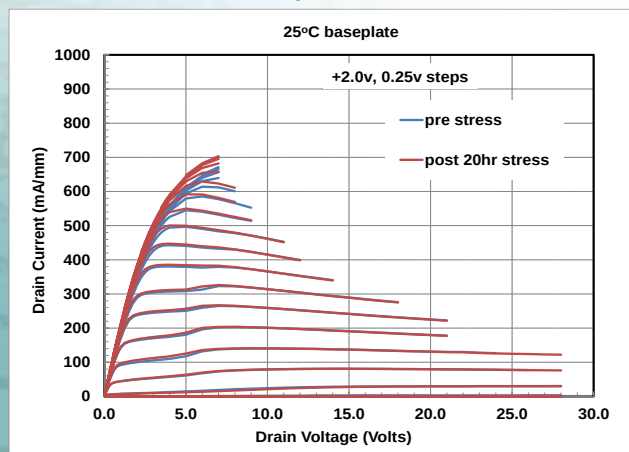
- Access to High Volume CMOS Supply Infrastructure
- High Volume Production Capacity
- Access to Advanced Packaging Technologies
- 150 mm Production Now - Roadmap to 200 mm

Accelerating GaN Technology into the Mainstream

- **Accomplishments**

- Established Au-free process which meets all DC parametric performance targets
- Demonstrated process on-wafer reliability
- Demonstrated Through Silicon Vias
- Established 150mm EPI supply
- Demonstrated RF Performance
 - Final optimization is process

- **Process Reliability: Pre/Post Stress**



- **Timeline**

- Process Freeze: October 2017
- Process Qualification: Q4CY17 -Q1CY18
- Product Qualification: Q1CY18 – Q2CY18

Agenda

- ✓ GaN update in general
- ✓ Status of CMOS fab transfer
- **Epi update**
 - GaN-on-Silicon Crystal Growth MOCVD Reactor Platform
 - Starting Silicon Substrate Material
 - GaN HEMT Epitaxial Design
 - Manufacturability – Epiwafer Characterization Trend Charts

IQE Production of GaN Epiwafers

MACOM



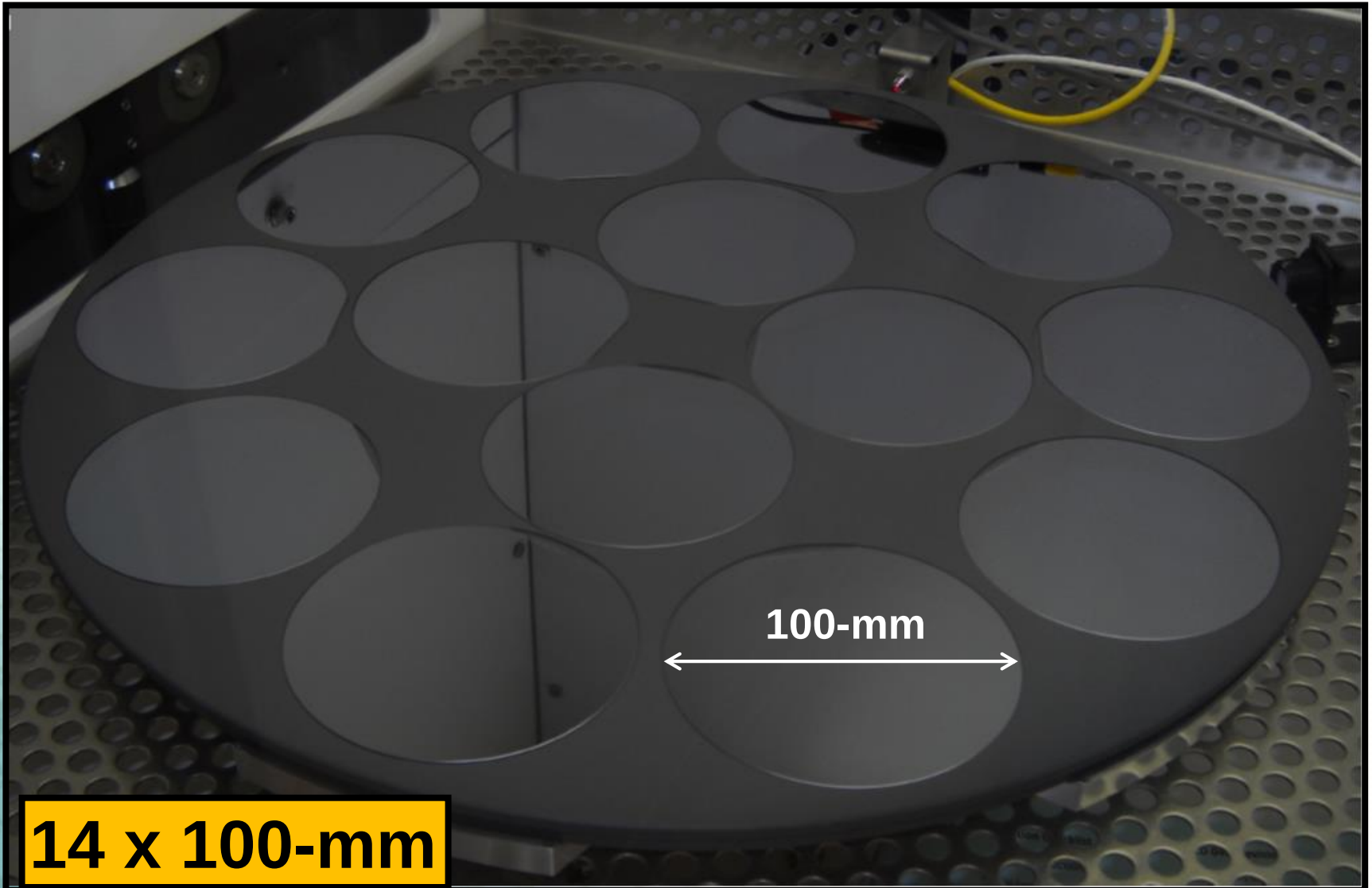
IQE GaN Epiwafer Manufacturing Facility
– Taunton, MA



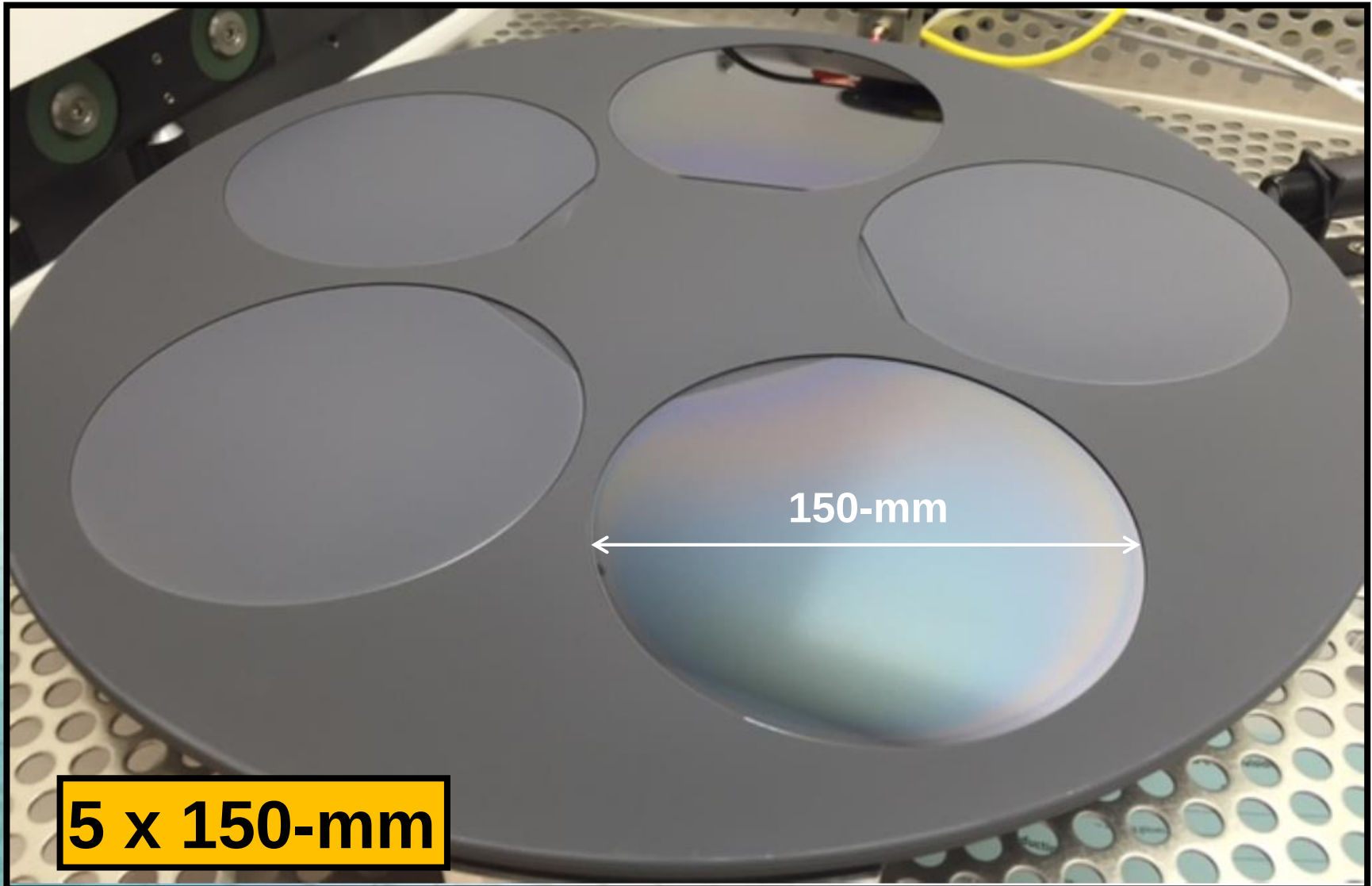
GaN-on-Silicon
Production Epi growth
- Veeco K465i
MOCVD Reactor



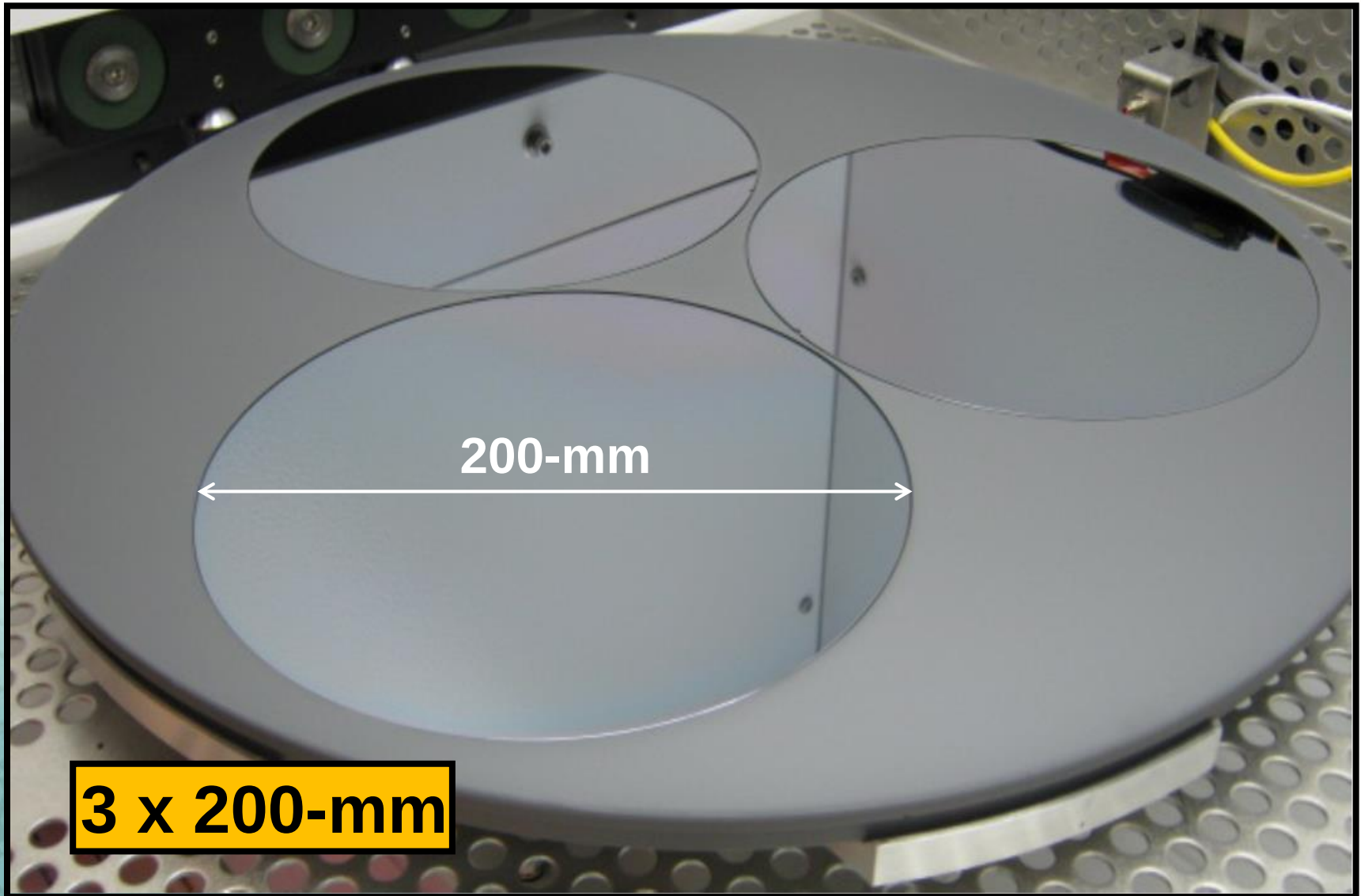
K465i MOCVD Production Reactor - Platter Configurations



K465i MOCVD Production Reactor - Platter Configurations



K465i MOCVD Production Reactor - Platter Configurations

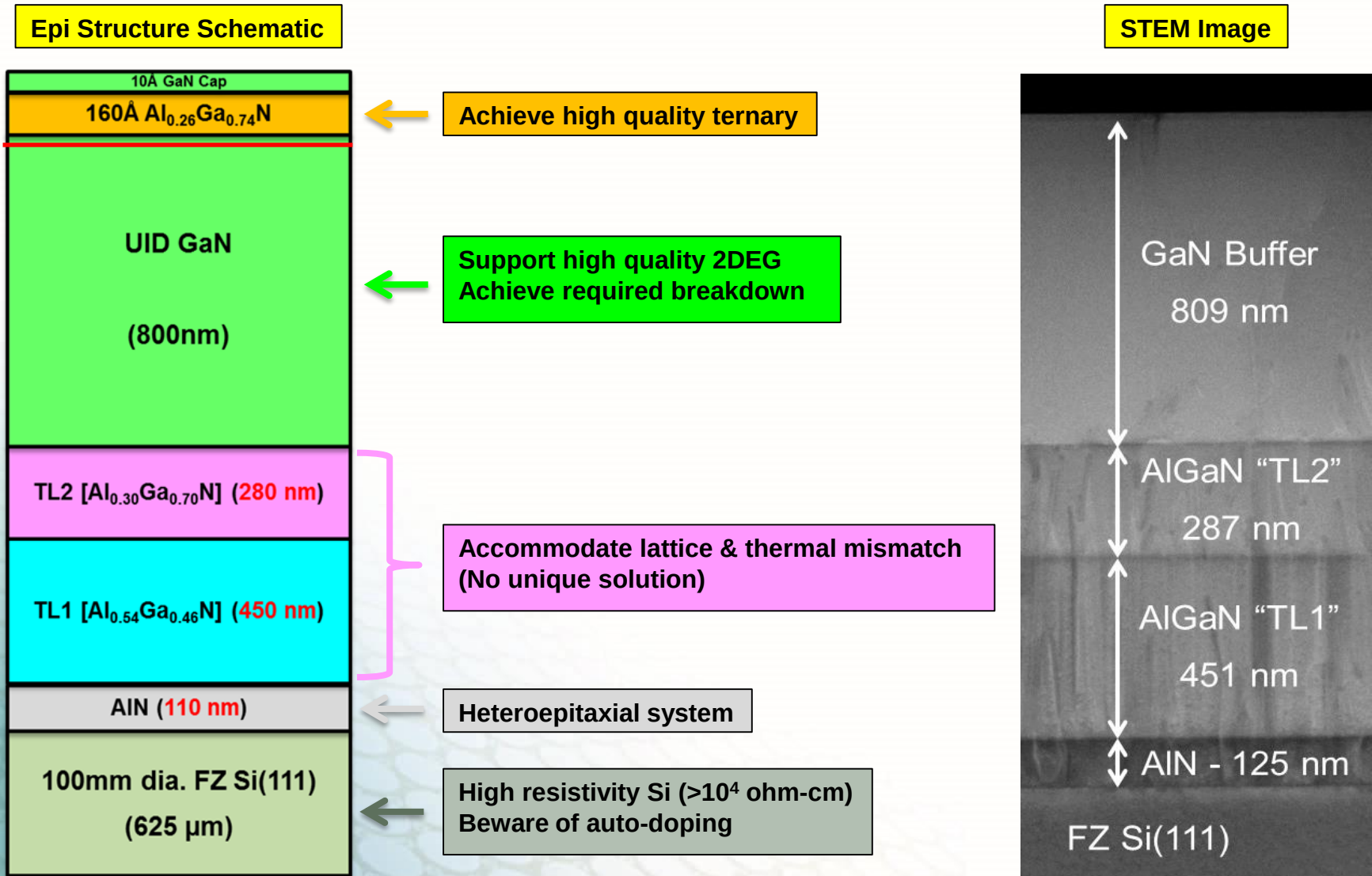


Silicon Substrates – Resistivity Availability vs. Diameter



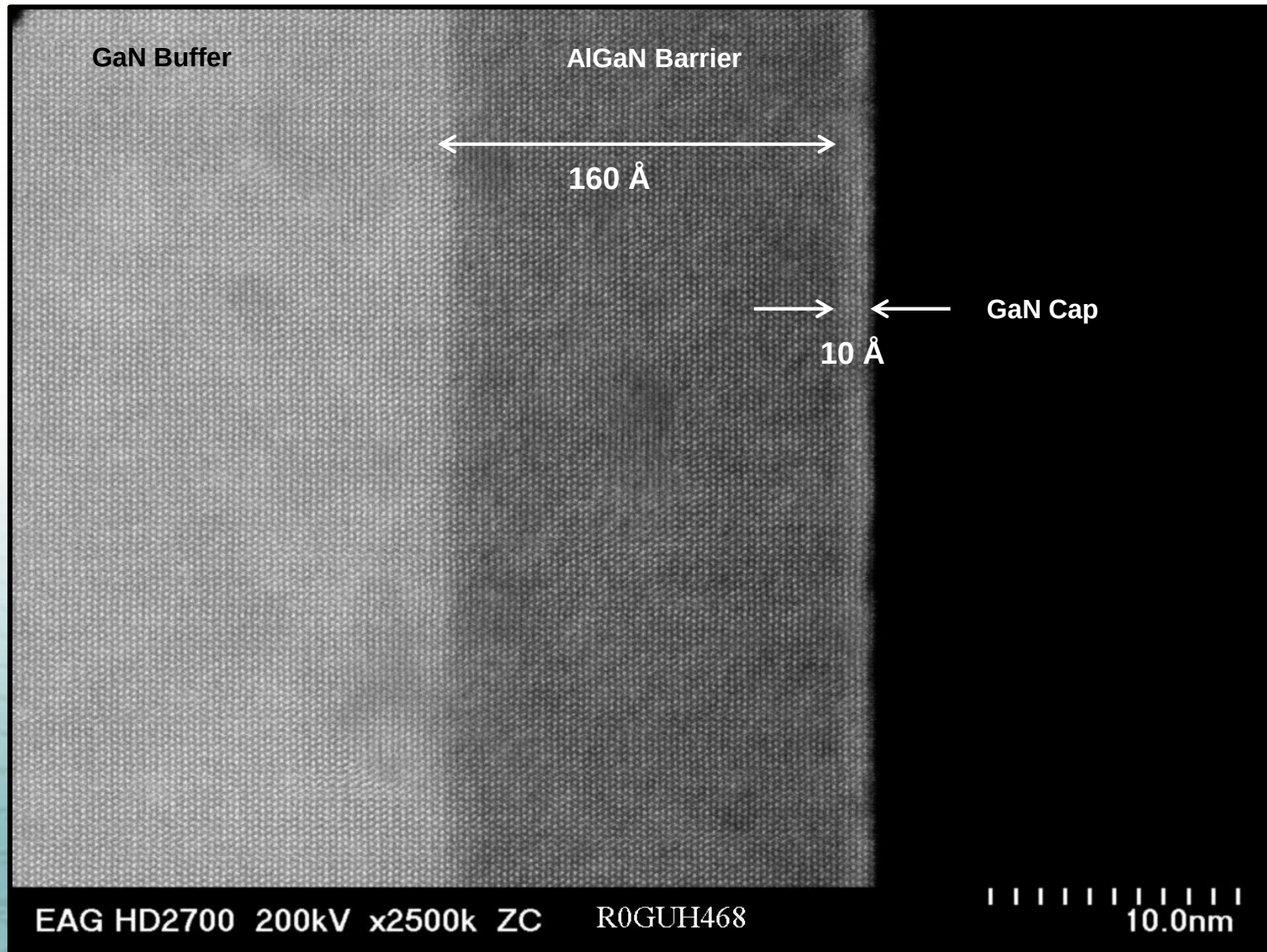
Wafer Dia. (mm)	Silicon Type	Orientation	Resistivity (ohm-cm)	Thickness (microns)	Flat Type	Foundry
100	Float Zone	(111)	$\geq 10,000$	625	Flat	MACOM or GCS
150	Float Zone	(111)	$\geq 3,000$	675	Notch or Flat	CMOS Fab
200	Mag. CZ	(111)	$\geq 3,000$	725	Notch	CMOS Fab
200	Float Zone	(111)	Development	675	Notch	CMOS Fab

Anatomy of a GaN-on-Si Epi Wafer

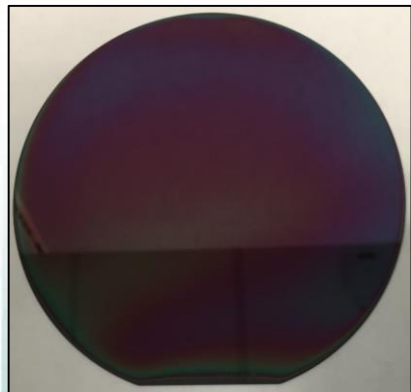


GaN HEMT: AlGaN Barrier Layer & GaN Cap

- TEM Micrograph



GaN-on-Si MOCVD Growth - Migration to Larger Diameter Wafers

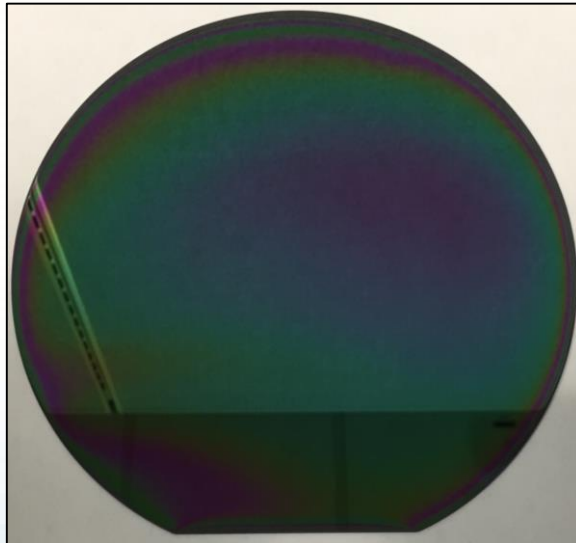


100 mm

Qualified

[PT-0008569]

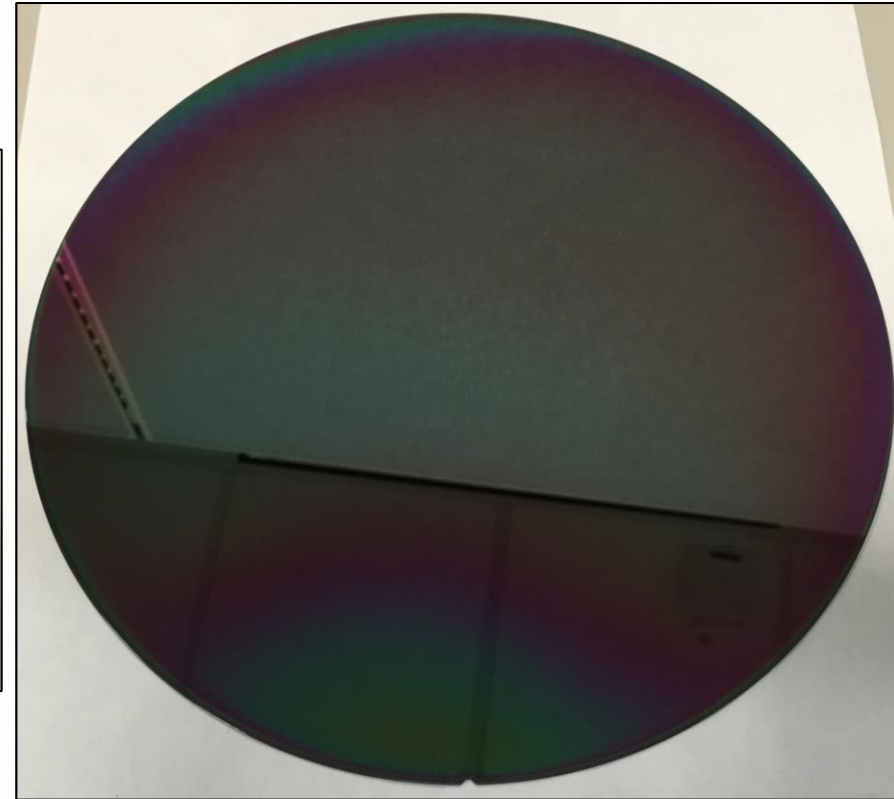
>1000 wafers
delivered.



150 mm

In Qualification

> 200 wafers
delivered.

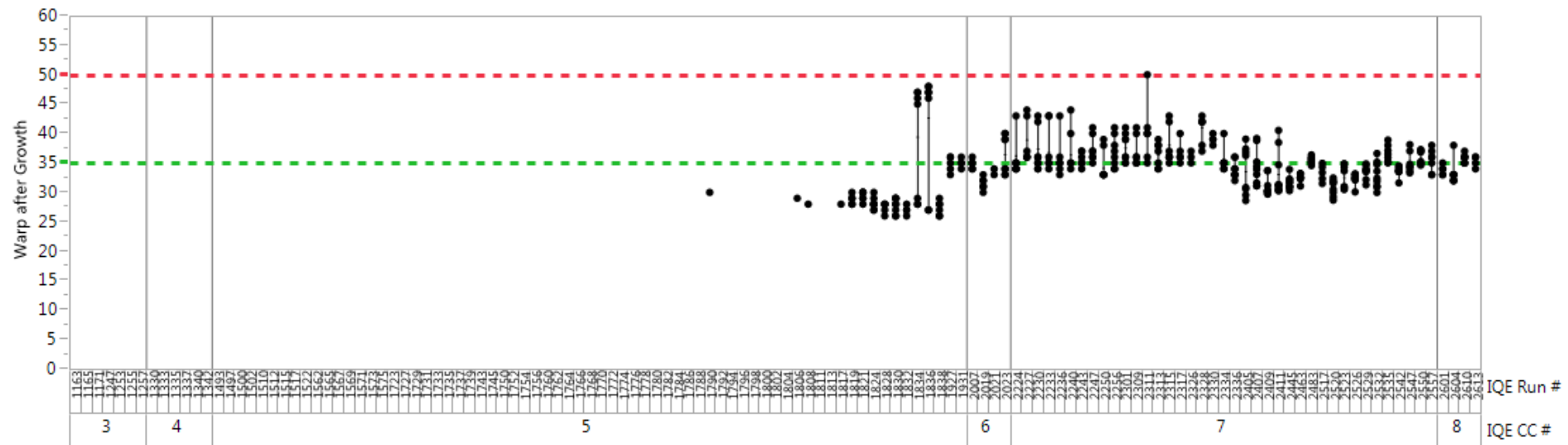
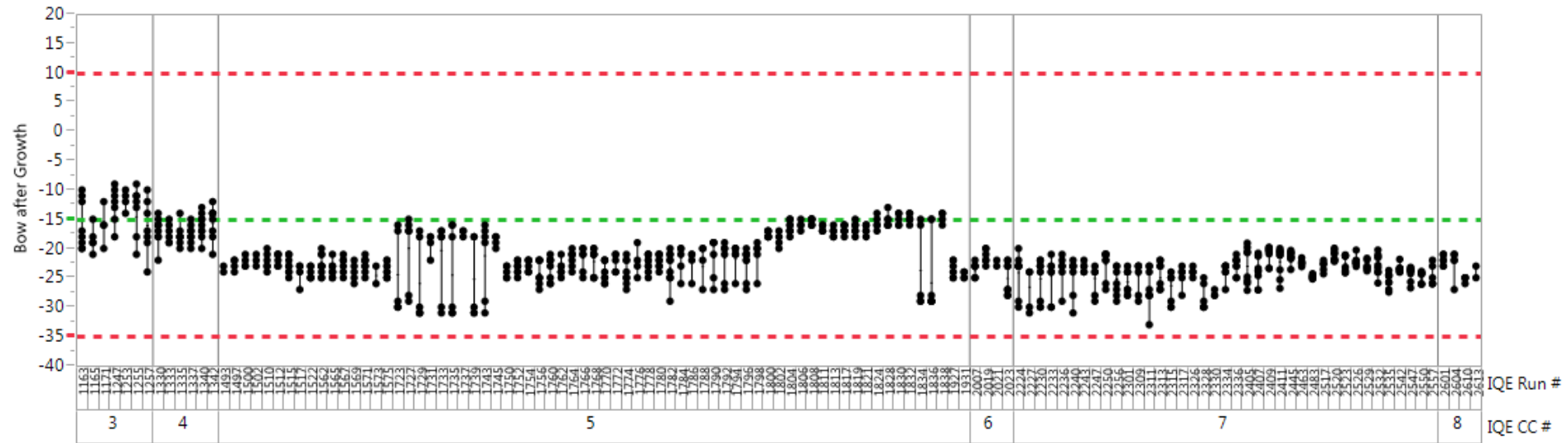


200 mm

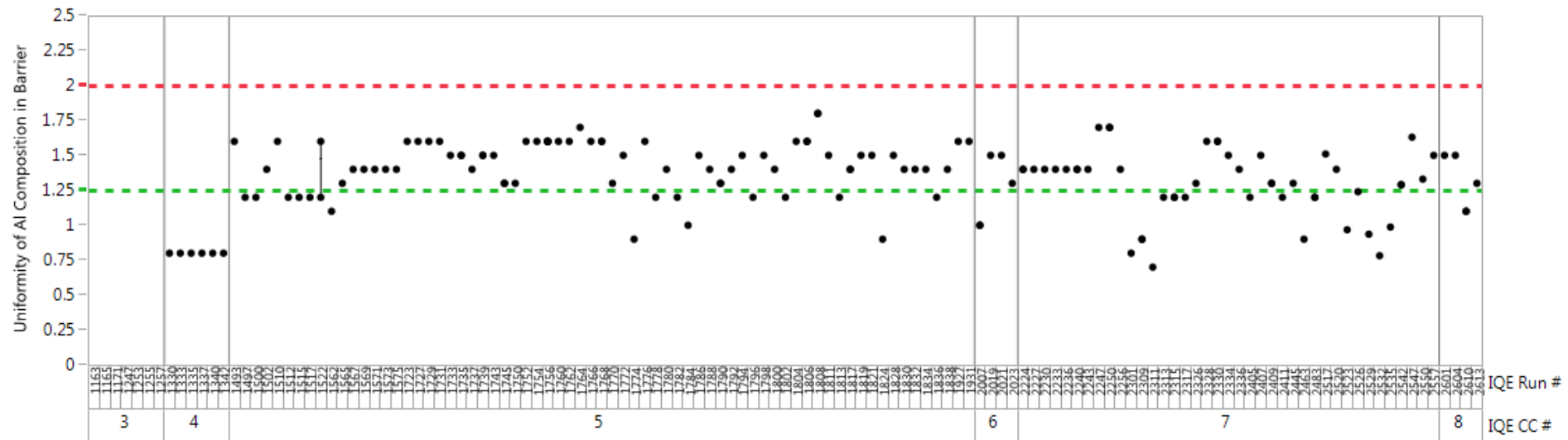
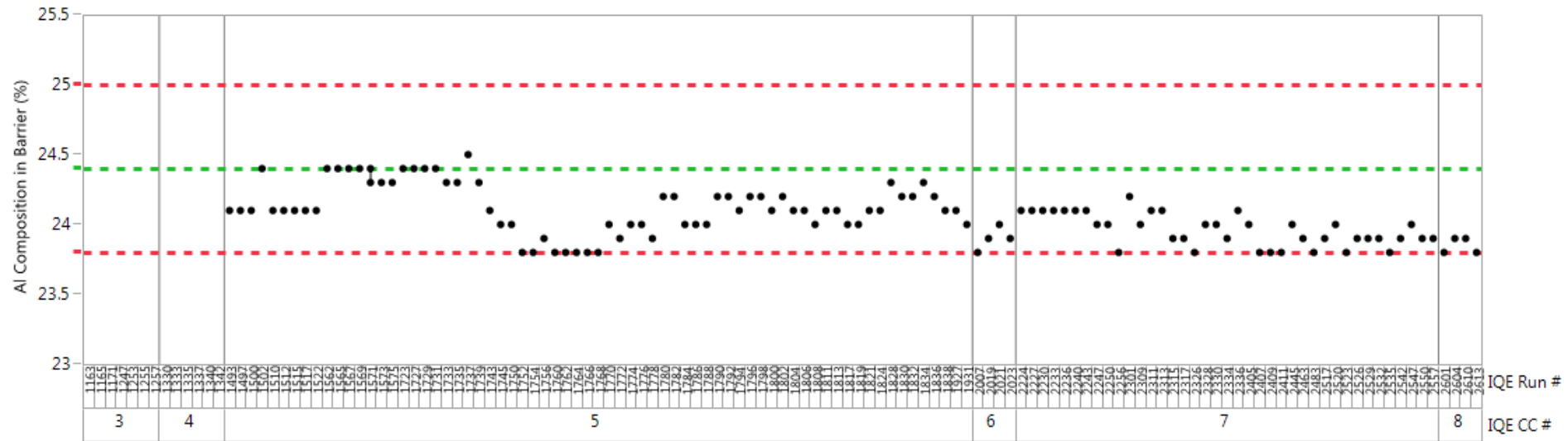
In development

Successfully demonstrated.

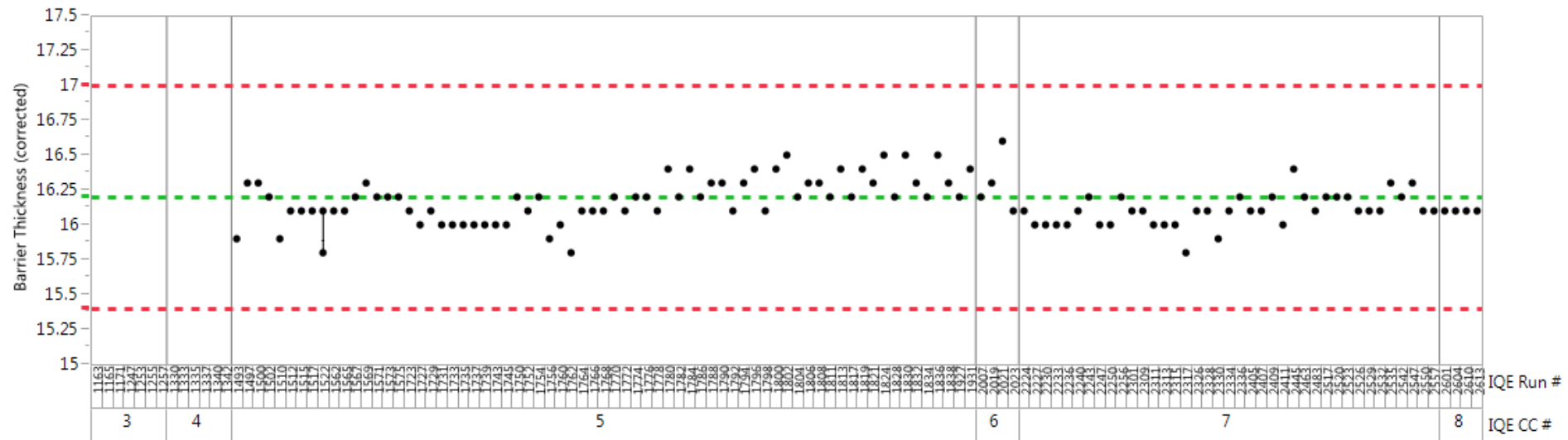
Total Wafer Bow & Warp (microns) [every wafer measured]



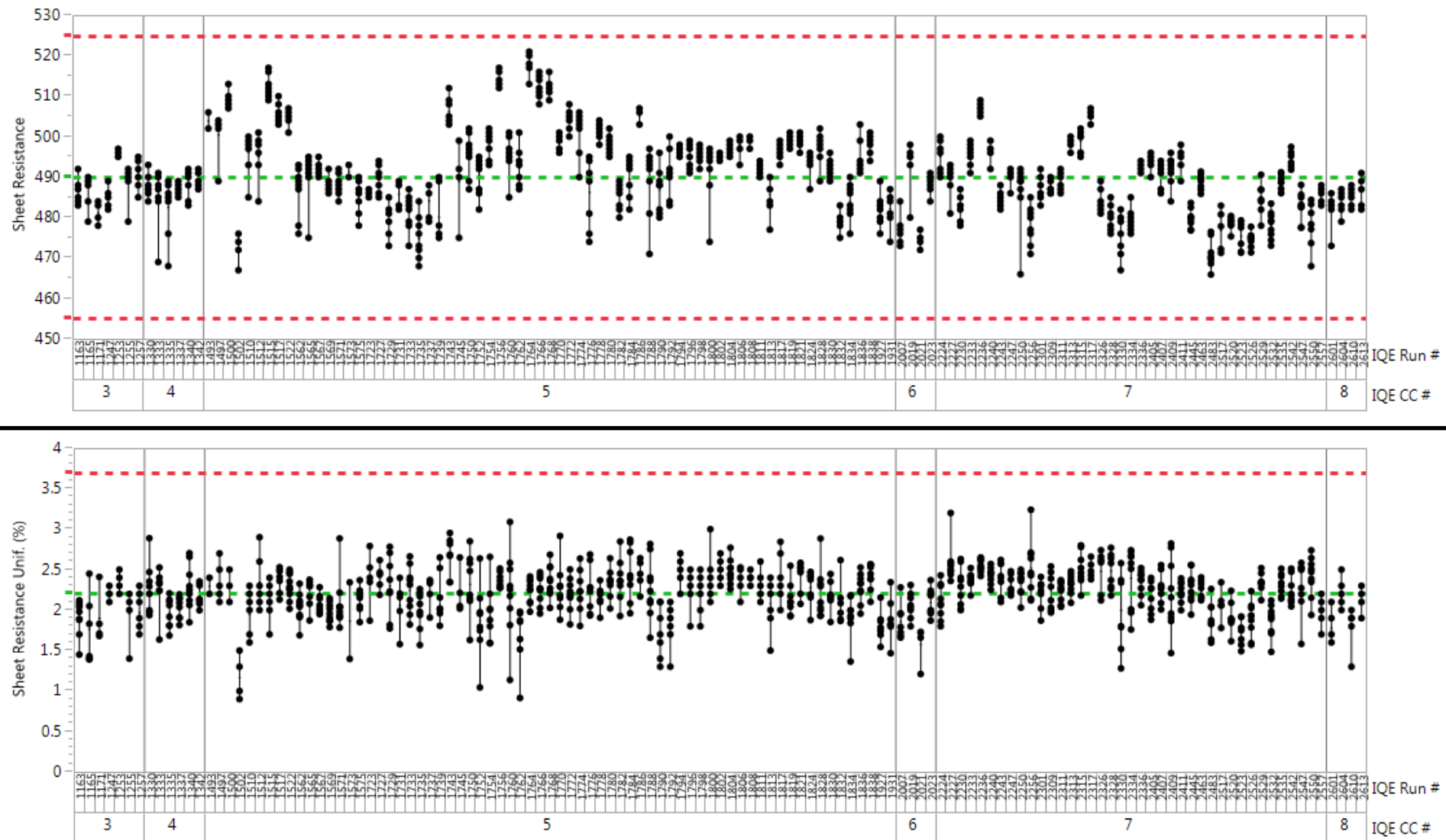
MACOM



AlGaN Barrier Thickness (nm) [one wafer per run measured]



Sheet Resistance (ohm/sq) & Uniformity [every wafer measured]



Agenda

- ✓ GaN update in general
- ✓ Status of CMOS fab transfer
- ✓ Epi update
- **Process Technology Corner lots**
- Reliability update – bHAST+
- Idq Drift
- Epiwafer technology development lab tour

In-Line Reliability Monitoring

Background / Motivation

- Process corner lots are used as one means of determining product sensitivity to key process / manufacturing variation
- Primary output – sensitivity / response functions for key product metrics
- Models are subsequently employed to refine in-line specification limits and inform yield improvement activities
- Experimental design is performed in a manner that permits follow-on lots to further refine understanding of other sources of variation and variable interaction

MAGB-100710-550S0S

Initial Corner Lot Investigation



Primary Variables:

- Primary variables: gate length, epitaxial mole fraction, barrier thickness
- Range of variation chosen at or near six sigma limits to assure response detection / sensitivity model fit (see supporting data)

Experimental Design:

- Two process lots, six wafers each, 180 builds
- Inclusion of inter- and intra-lot replicates to improve ability to account for other sources of variation
- Design structured to facilitate follow-on lot options to refine model for interactions / other sources of variation

Primary Response:

- All primary product specifications, KGD yields

Mask Set 2220 Engineering Rev X _

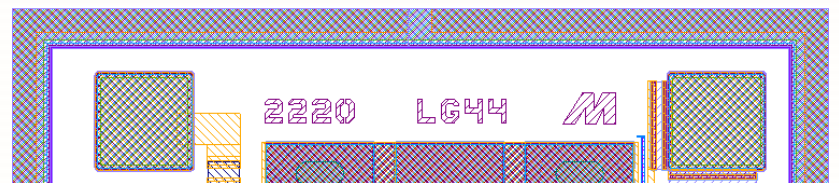
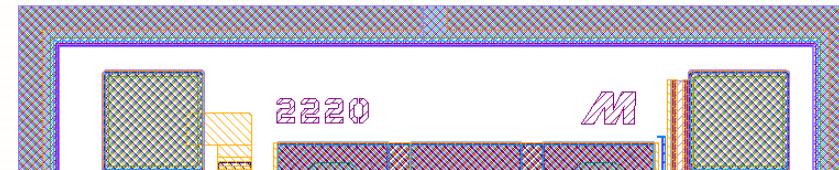
Varying Gate Length



$L_g = 0.36 \mu\text{m}$

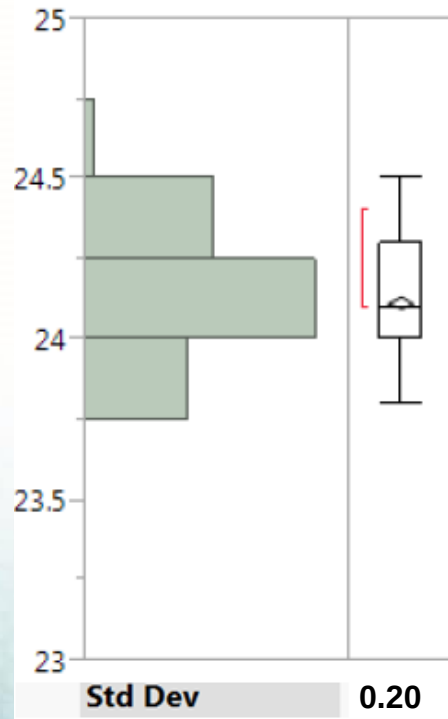
$L_g = 0.40 \mu\text{m}$

$L_g = 0.44 \mu\text{m}$



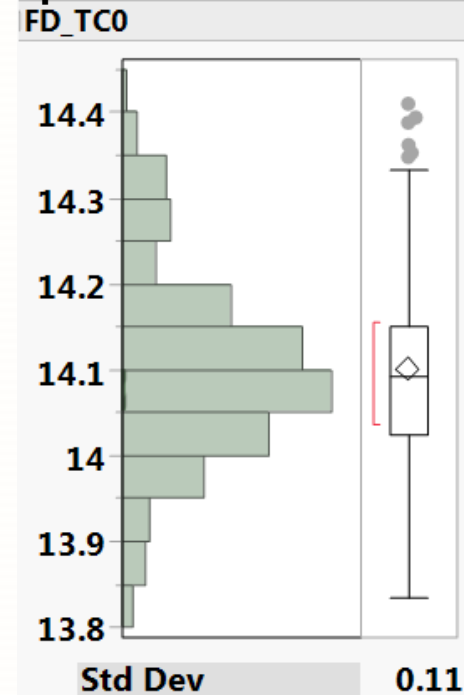
- Gate variation is established on each of the wafers by use of experimental gate reticle
- Die test is coded to indicate experimental variant
- Actual gate lengths are validated by CD SEM test plan with extended coverage

Supporting Data: Al% and TC0 Variations – Measurement Data of Baseline Epi



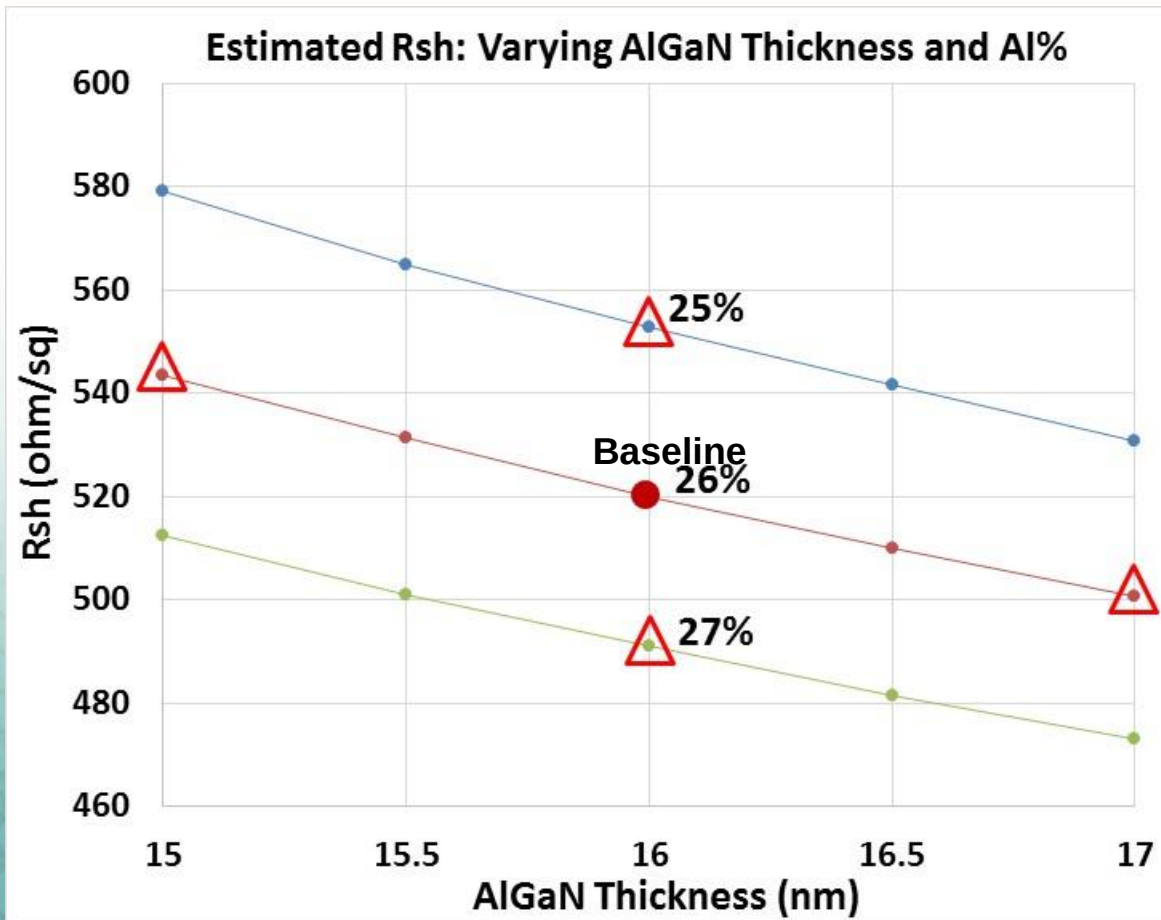
Epitaxial vendor data; $6 \cdot \sigma = 1.2\%$
→ Varying $\pm 1\%$ around the baseline
in the experiment

Equivalent AlGaIn thickness



MACOM PCM data; $6 \cdot \sigma = 0.66\text{nm}$
→ Varying $\pm 1\text{nm}$ around the baseline
in the experiment

Supporting Data: Estimated Rsh



Rsh of the baseline structure is from PCM measurement. Rsh of other structures is estimated by comparing the calculated charge density against the baseline and assuming a constant mobility.

$\text{Al}_x\text{Ga}_{1-x}\text{N}$ Barrier Variants

Target Thickness (nm)	AlN mole fraction, x	Estimated Rsh (ohm/sq)	MACOM Epi Code	Production Epi No.
16	0.25	553	MTL-23	N/A
15	0.26	543	MTL-21	N/A
16		520	MTL-17	PT-0008569
17		501	MTL-22	N/A
16	0.27	491	MTL-24	N/A

All other layers should be the same as MTL-17/PT-0008569.

Agenda

- ✓ GaN update in general
- ✓ Status of CMOS fab transfer
- ✓ Epi update
- ✓ Process Corner lots
- **Reliability update – bHAST+**
 - Process Reliability Plan
 - bHAST results-to-date
 - THB results-to-date

Process Reliability Plan

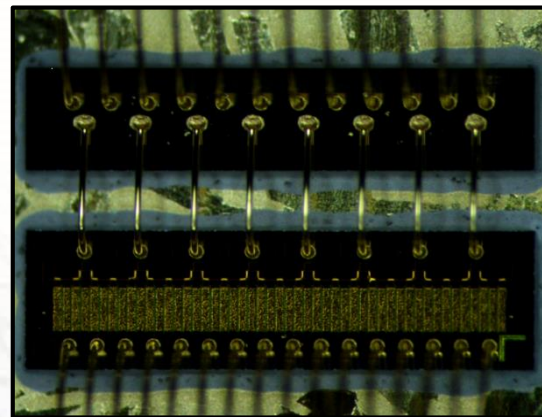
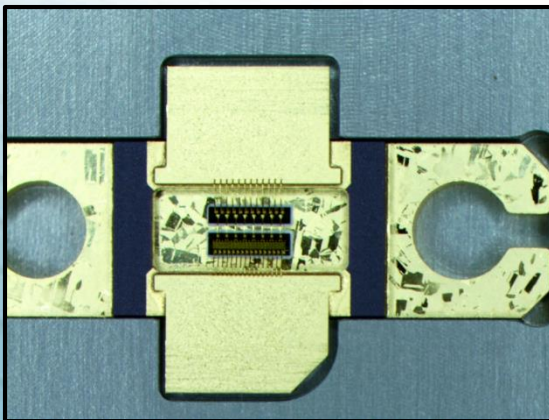


Test	Condition	End Point	Sample Size	Reliability Target
Accelerated Life Test (ALT): DC biased tests	Test structure: 2mm discrete FET Devices from 3 wafer lots stressed at 3 channel temperatures (e.g. 290, 310, 330 C) Vdd=50V; Id=80 mA/mm;	Failure criteria: 1 dB drop in RF output power (fixed input power level) 20% drop in Idss 3-terminal off leakage > 1mA/mm - Monitor other parameters; consider any "out-of-family" changes	DC: 12 samples per temperature, per lot	MTTF>10 ⁶ hrs @ 225°C (based on modeled channel temperature)
HAST or THB (Not required for hermetic packages)	Biased HAST: 96 hours, 85%RH, 130C, Vds = 50V, Vgs = -8V (in accordance with JEDEC JESD22-A110)	Same as ALT May choose to run THB (85/85, 1,000 hrs) as well	15 per lot x 3 lots; Non-hermetic assemblies (open top assemblies)	No Failures - Also plan tests with nominal Pdis - Tests planned to establish acceleration model
Temp Cycle	JESD22-A104, cond. C: 500 Thermal Cycles of -65 to +150 C, 10 minute dwell; die attached to an appropriate package or substrate	Electrical Test/Visual Inspection (catastrophic failure)	15 per lot x 3 lots	No Failures

Gen4 bHAST Assessments



- Test methodology employed is described in JESD22-A110-B
- Environmental Conditions: 85% Relative Humidity, 130C Temperature
- Bias Conditions: $V_{dd} = 50V$, $V_{gd} = -6V$, $I_{dd} = 0A$
- Test Duration: 96hrs.
- Failure Criteria: Catastrophic Failures of Exceeding Parametric Spec Limits
- bHAST Qualification Test Vehicle: MAGB-100025-060BOS
 - AC360 Non Hermetic Air Cavity Ceramic Package
 - Single 10mm gate periphery GaN die and pre-match capacitor with Ag epoxy die attach
 - This qual vehicle was chosen as it is representative of all our non hermetic products (i.e., Gen4 GaN die, pre-match capacitor, Ag epoxy die attach, non hermetic package).



Gen4 bHAST Assessments



- The Gen4 Process Qualification plan is our first GaN product attempt to meet the stringent requirements of passing bHAST or THB.
- As such, we have gone through over 22 different bHAST or THB evaluations with various degrees of success and have made several robustness improvements to the transistor process, die layout and assemblies as a result of our FA analysis.
- We have found that through the use of a combination of dielectric and organic passivation layers at both the wafer level and assembly level, we can achieve very high bHAST (130C / 85%RH / 96 hrs) reliability levels per the table below.

Lot ID	# Samples	# Catastrophic failures	# Parametric Limit failures	% Passing
2141001	10	0	0	100
2141011	30	0	0	100
2141012	22	0	0	100
Total	62	0	0	100

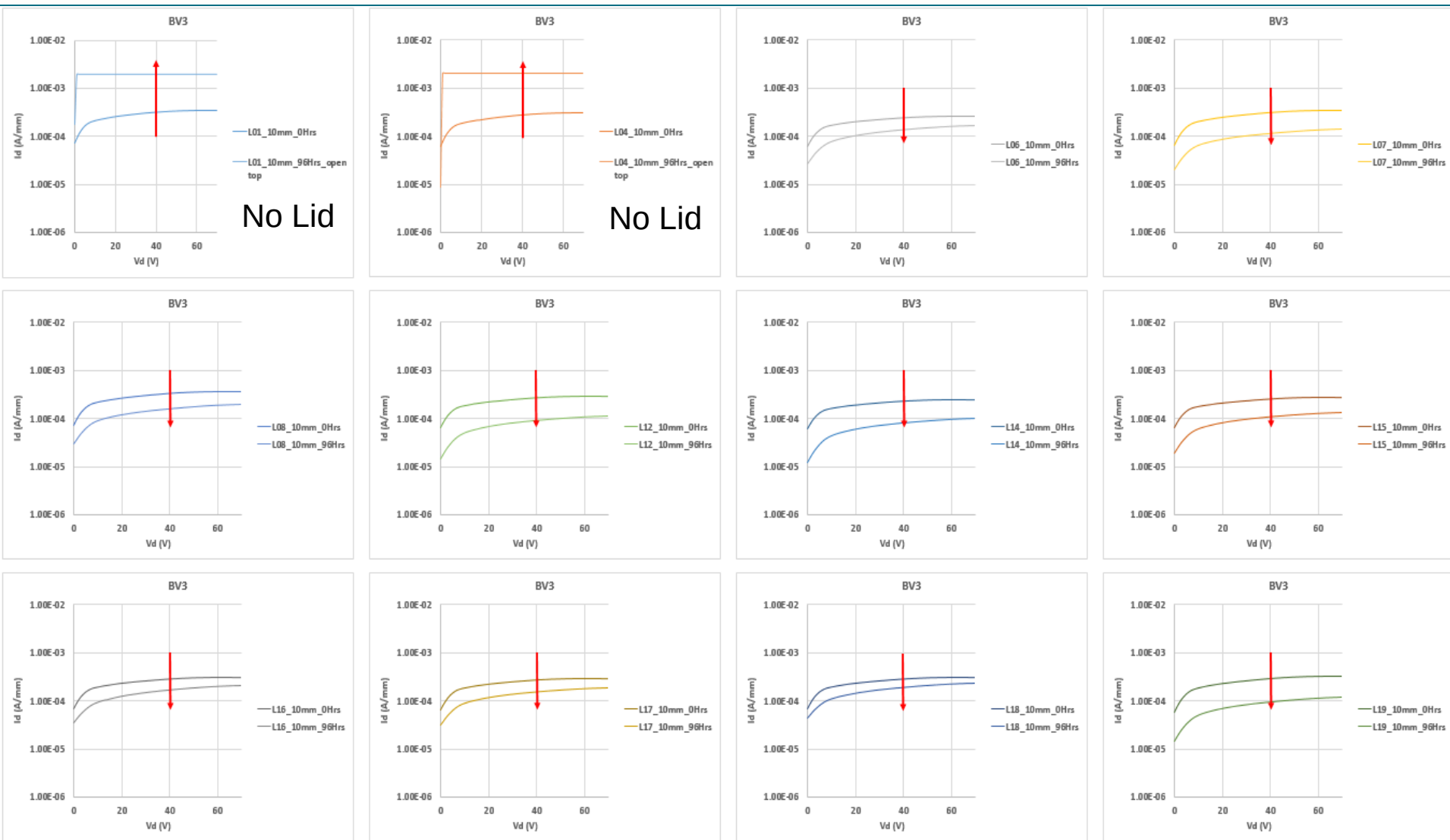
Gen4 bHAST Next Steps



- Increase sample size to exceed >15 parts per Nokia QRS document.
- Collect data on Parametric Drift on both DC and RF as a result of the Accelerated Lifetime Test.
- Increase HAST test time from 96hr to 198hrs per Nokia QRS.
- Discussion:
 - Per JEDEC, accelerated life tests are design to accelerate end of life. Although we are meeting the JEDEC defined success criteria, we would like to understand Nokia's thoughts on allowable "parametric drift" and your success criteria as this is not defined in the Nokia QRS document.
 - Additionally, per the Nokia QRS, it requires 5 devices to be de-lidded and a visual inspection for corrosion to be performed. Can you discuss the motivation here and share with us your success criteria for this inspection?

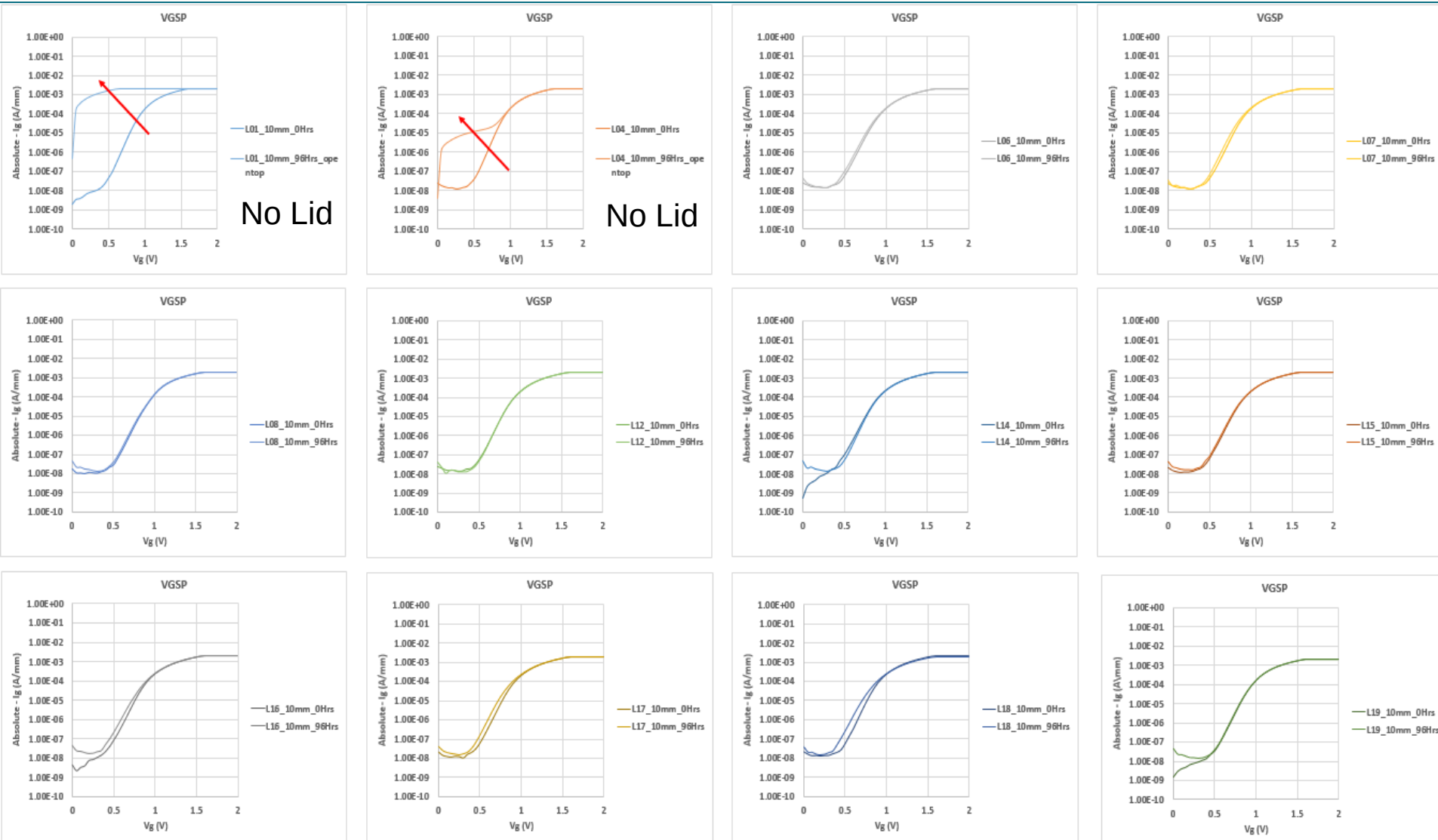
SPA – BV3 Curve on Board

0Hr vs. 96Hrs (HAST) lot 2141001



SPA – VGSP Curve on Board

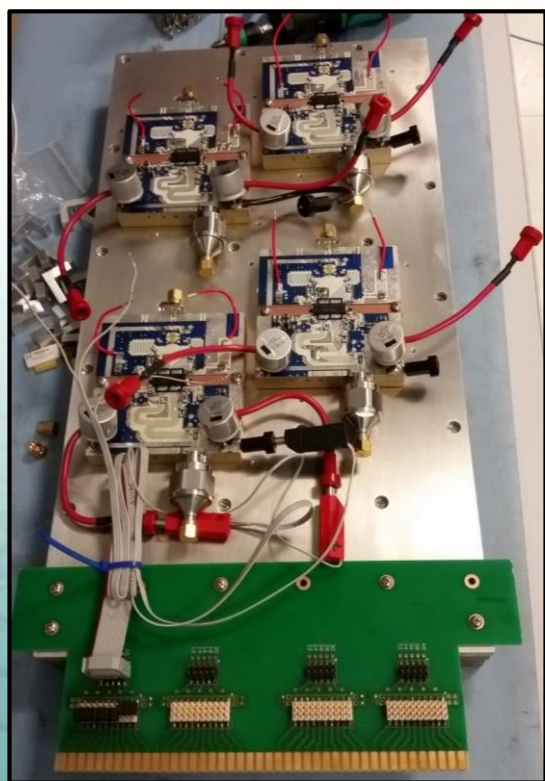
0Hr vs. 96Hrs (HAST) lot 2141001



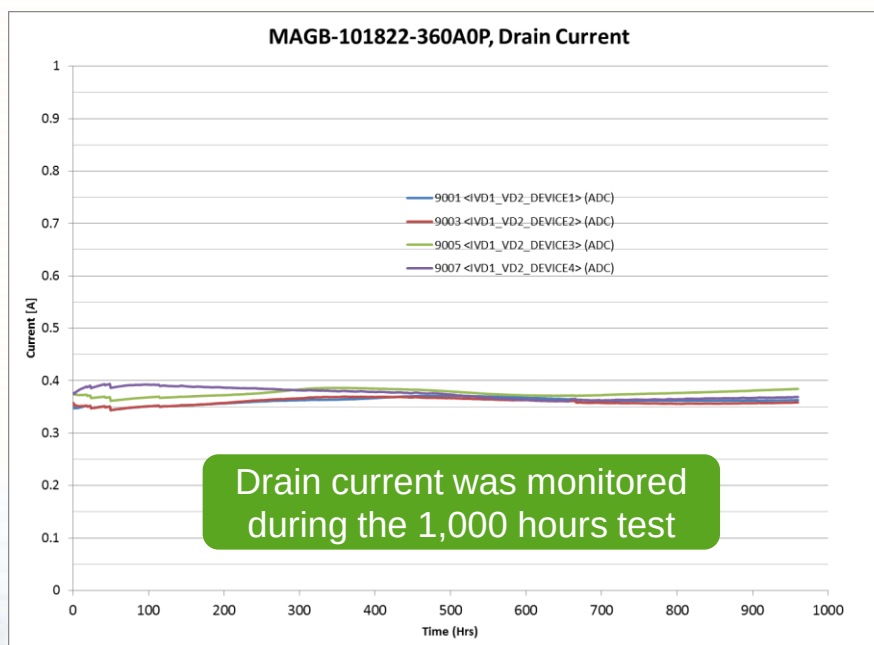
MAGB-101822-360A0P

THB (85°C / 85% RH, 1,000 hours)

160W + 200W, TO-272-4



4 Parts Tested



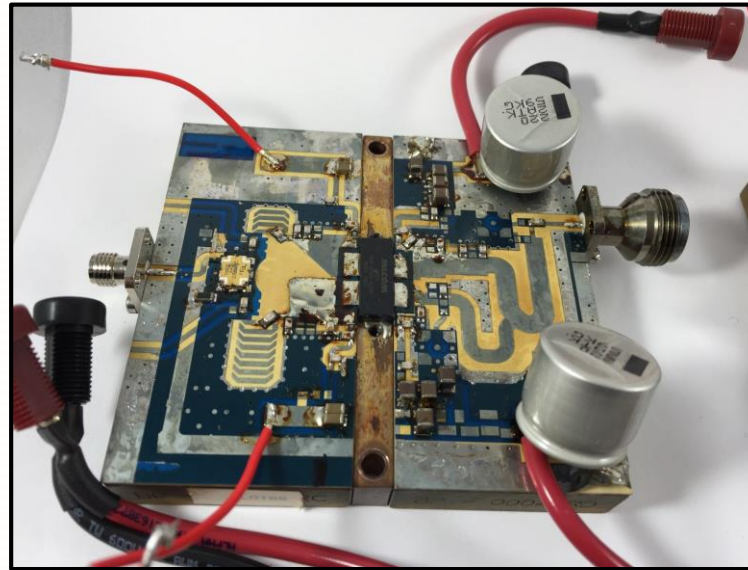
1000 Hrs THB Conclusion:

- No Change in drain current during the test
- RF test before and after show no change

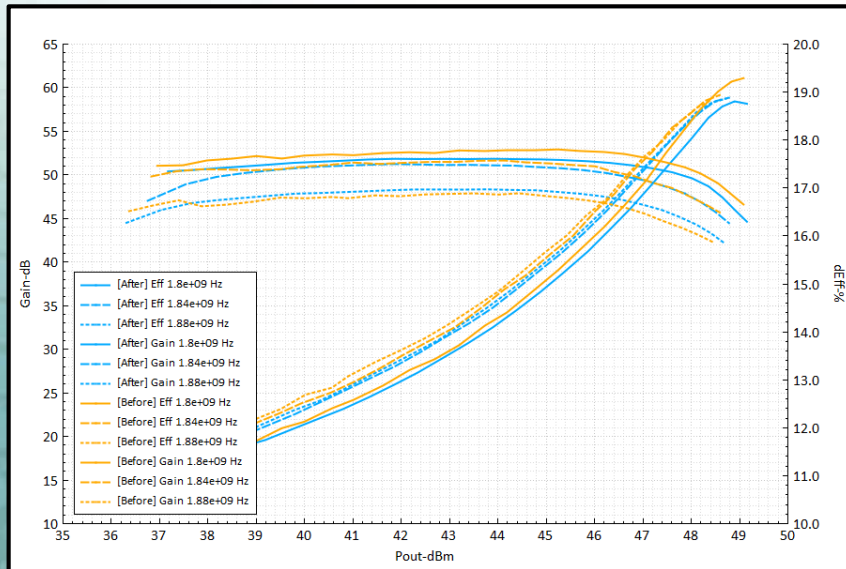
Electrical Conditions	Peaking	Carrier
Drain Voltage	50V	50V
Drain Current	0A	0.38A
RF Power	No RF	No RF

LL0166

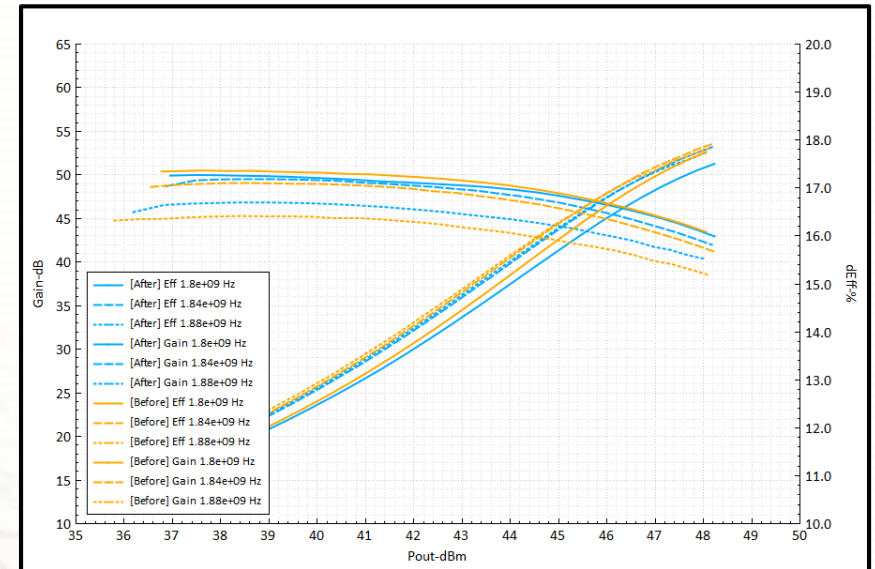
Demo after 1000 Hrs THB



Pulse CW 500us/ 10%

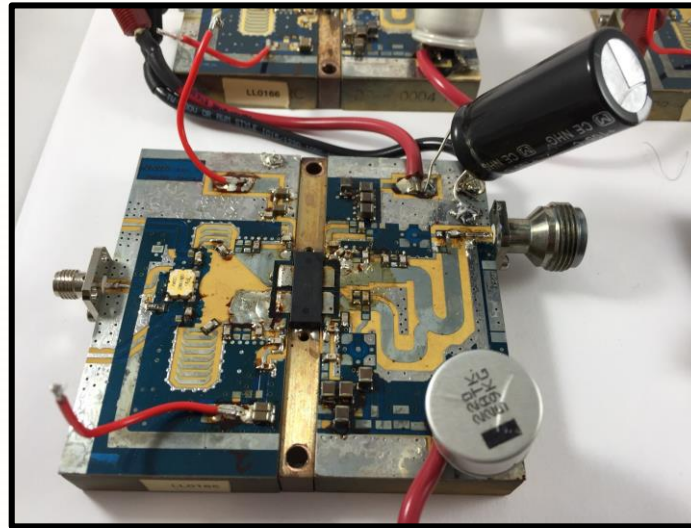


1C LTE 20 MHz 7.5dB

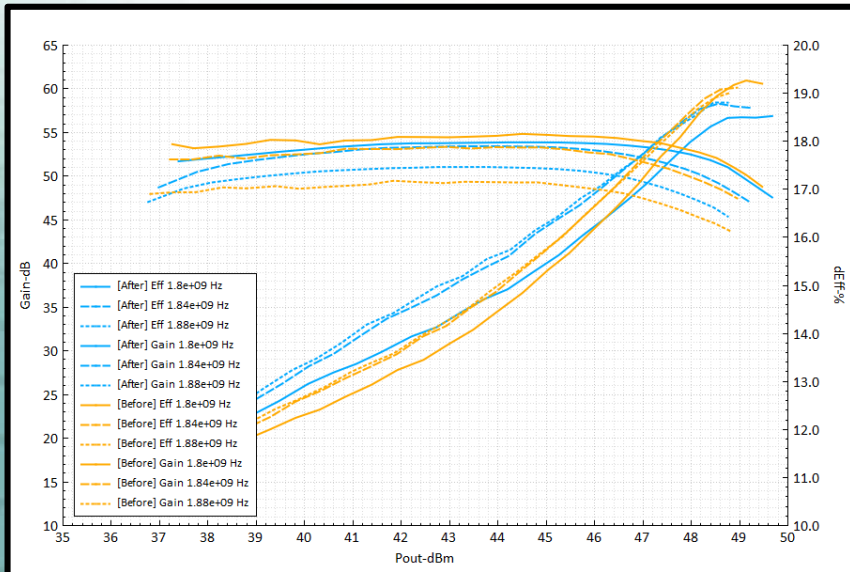


LL0165

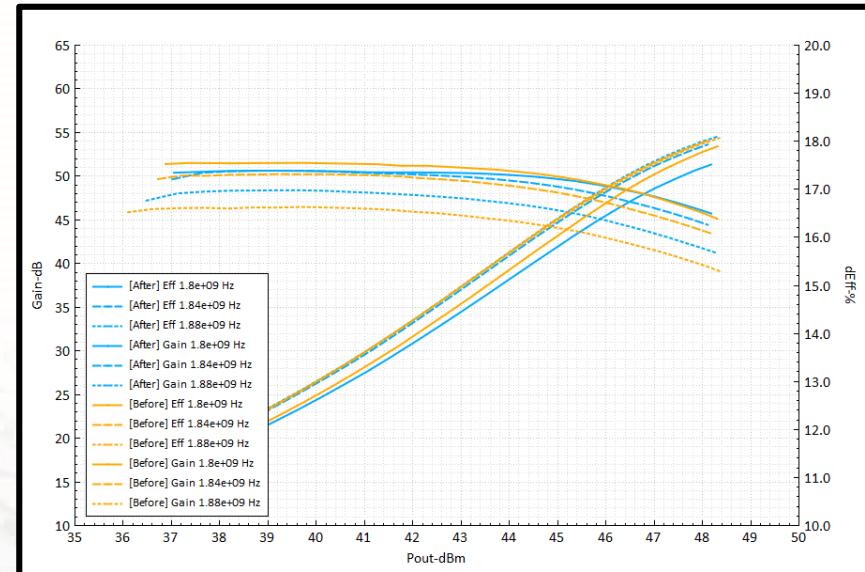
Demo after 1000 Hrs THB



Pulse CW 500us/ 10%



1C LTE 20 MHz 7.5dB

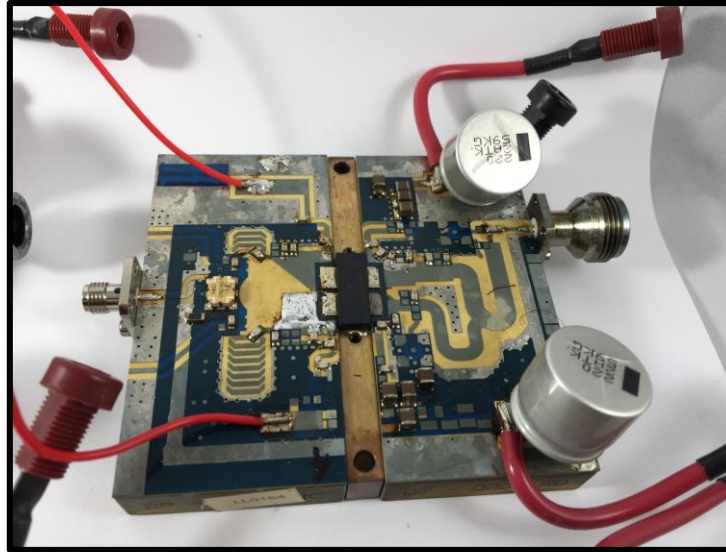


MACOM Proprietary, Shared with Nokia on 10 May 2017 under strict NDA

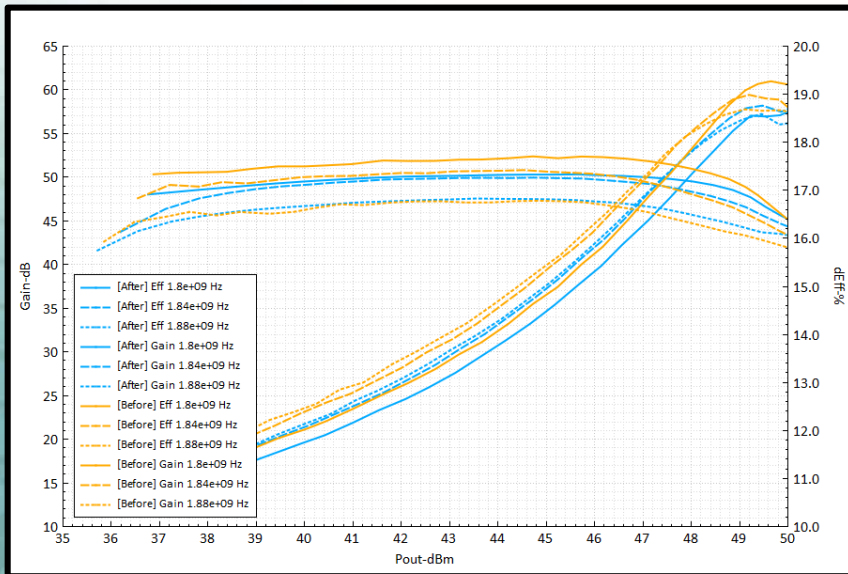
LL0164

Demo after 1000 Hrs THB

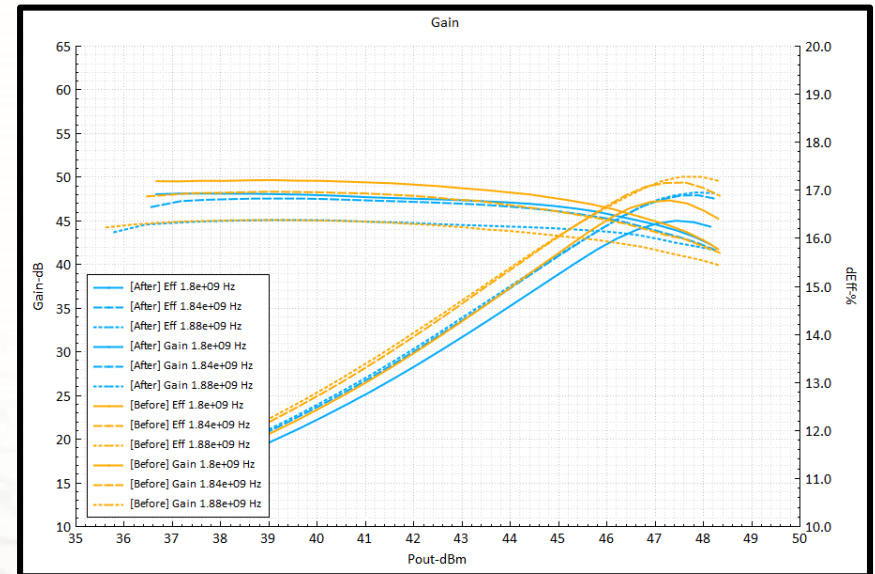
MACOM



Pulse CW 500us/ 10%



1C LTE 20 MHz 7.5dB

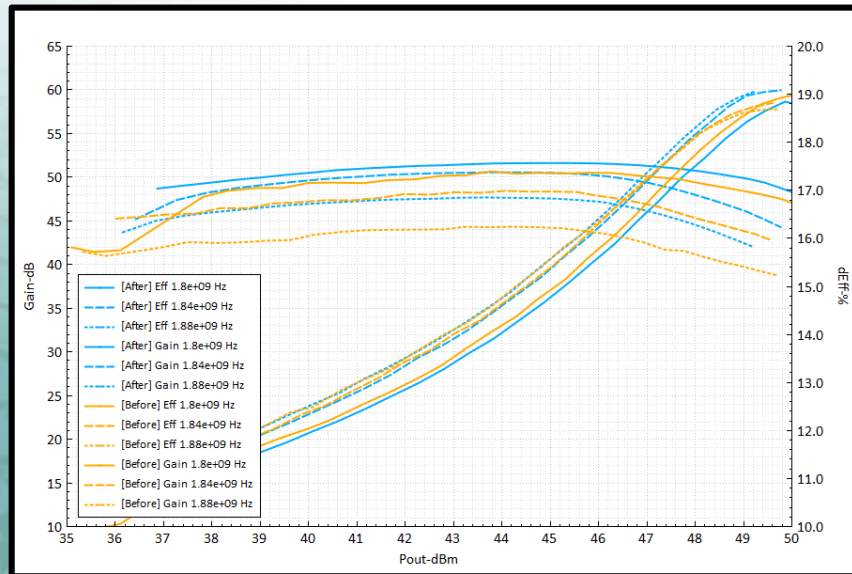


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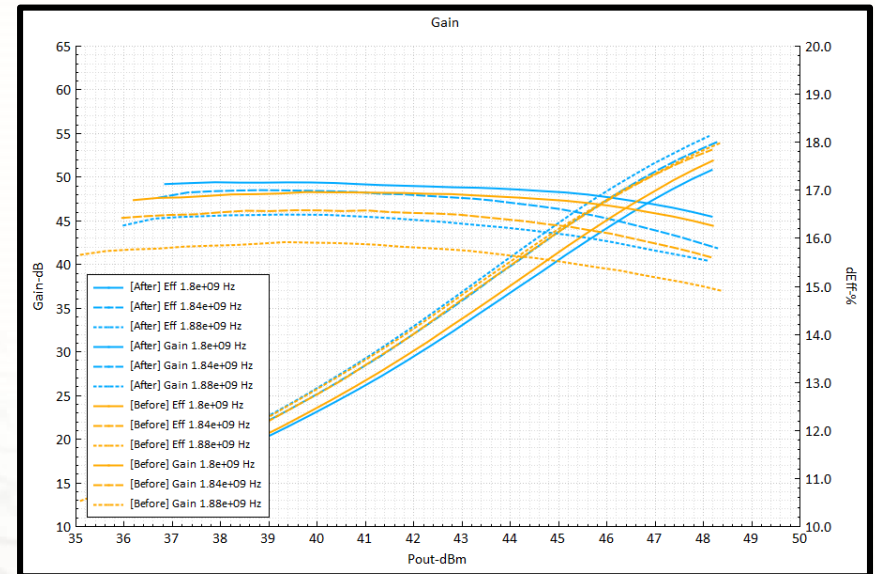
Demo after 1000 Hrs THB



Pulse CW 500us/ 10%



1C LTE 20 MHz 7.5dB



MAGB-100710-550S0S

THB (85°C / 85% RH, 1,000 hours)

280W + 280W, AC780



250 Hours so far with no
change in drain current



Electrical Conditions	Peaking	Carrier
Drain Voltage	50V	50V
Drain Current	0A	0.4A
RF Power	No RF	No RF

Agenda

- ✓ GaN update in general
- ✓ Status of CMOS fab transfer
- ✓ Epi update
- ✓ Process Corner lots
- ✓ Reliability update – bHAST+
- **Idq Drift**
- Epiwafer technology development lab tour

IDQ drift devices under test



- $W_g = 12 \times 170 \mu\text{m} = 2.04 \text{ mm}$
- Starting $P_{\text{diss}} = 1.92 \text{ W} = 0.925 \text{ W/mm}$
 - $V_{\text{DS}} = +50 \text{ V}$
 - Set V_{GS} to obtain $I_D = 38 \text{ mA} = 19 \text{ mA/mm}$ (+/- 1.5 mA/mm)
 - Keep V_{GS} constant at above value (do not adjust to maintain $I_D = 38 \text{ mA}$)
- $T_b = 195 \text{ }^\circ\text{C} \rightarrow$ starting $T_j = 225 \text{ }^\circ\text{C}$ (product max operating T)
- 6 parts started (sampling both GCS and MACOM Lowell Foundries)
- Monitor drain current at constant V_{GS}

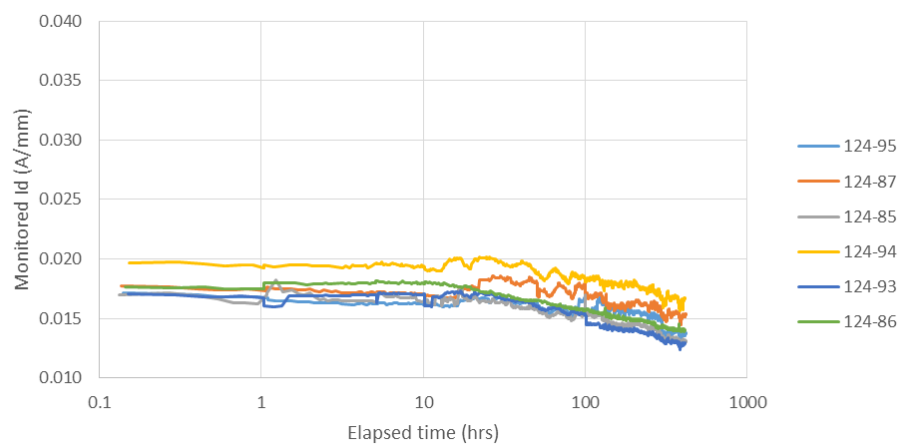
Monitored drain current



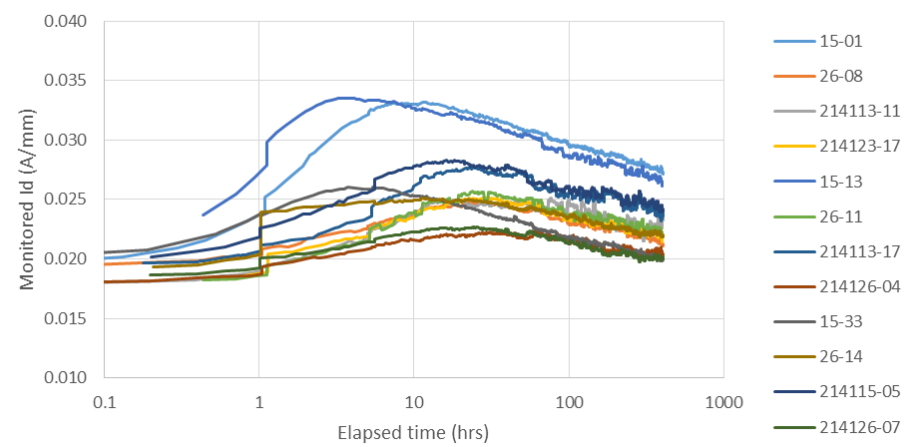
GCS

MACOM

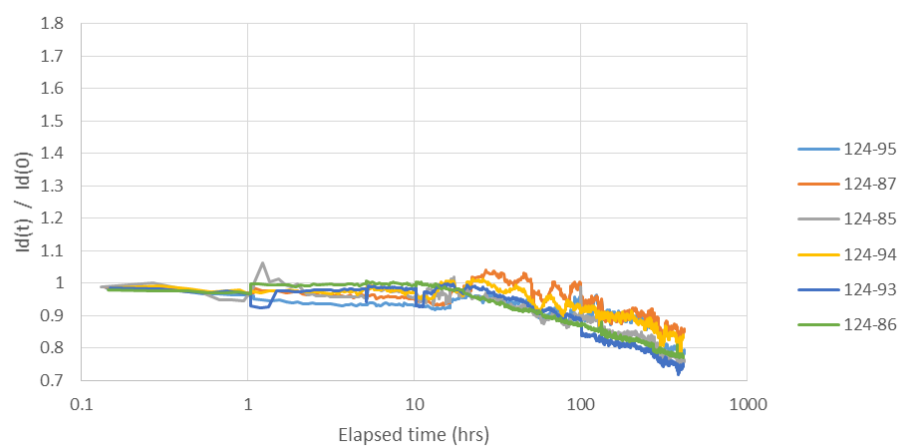
Monitored Id throughout stress test



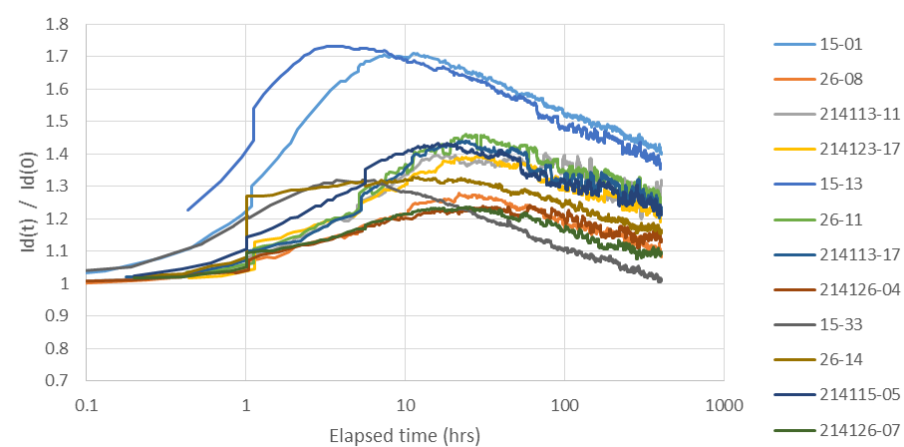
Monitored Id throughout stress test



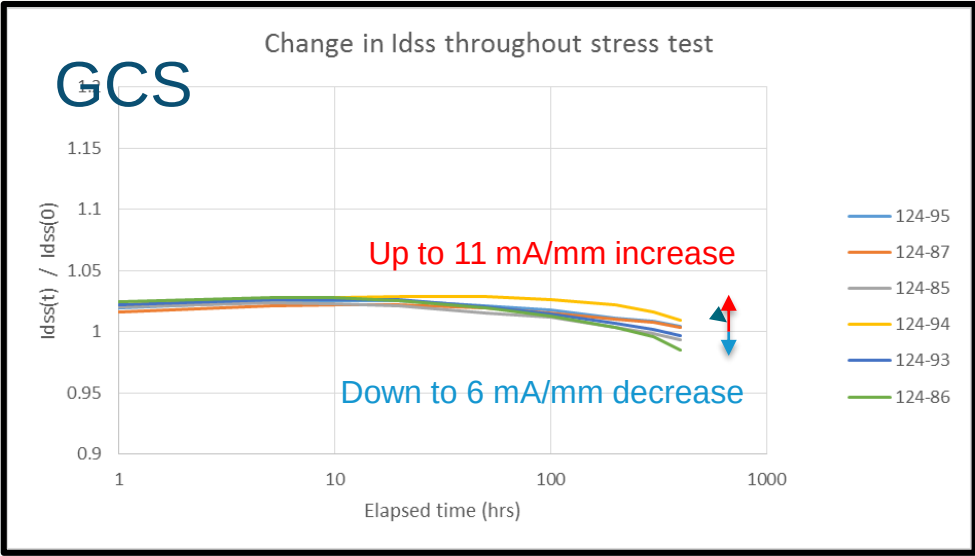
Change in monitored Id throughout stress test



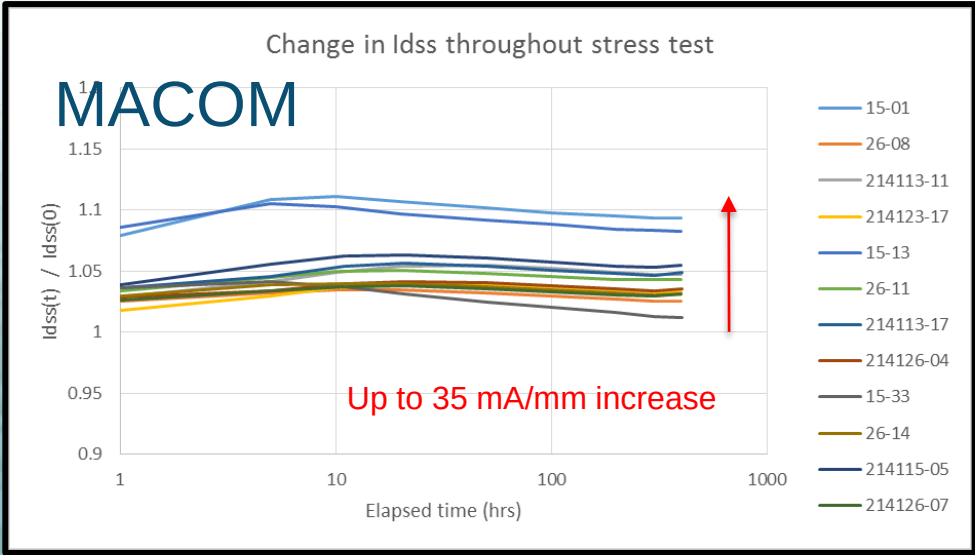
Change in monitored Id throughout stress test



Change in Idss degradation



XFER curve
VDS = +10 V
Sweep VGS from -4 V to +1 V
Define IDSS as ID @ VGS = 0 V

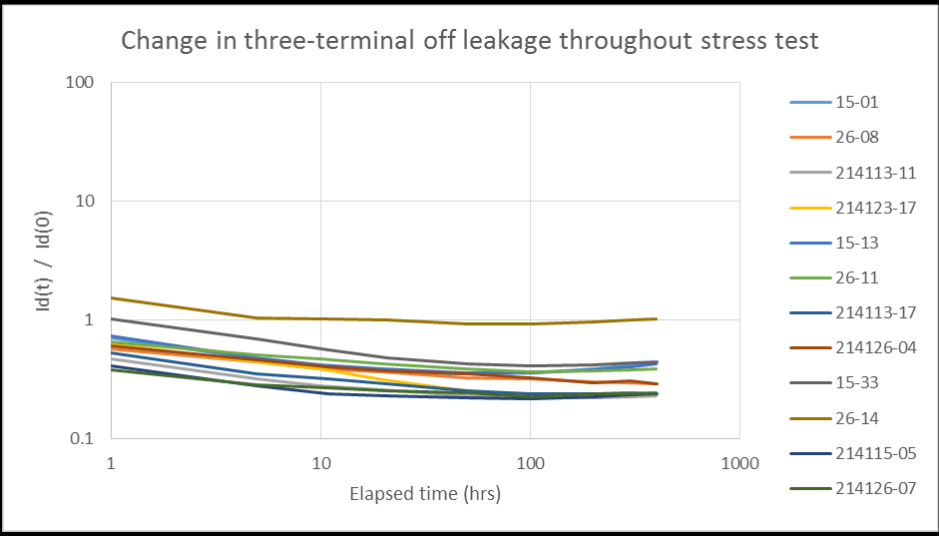
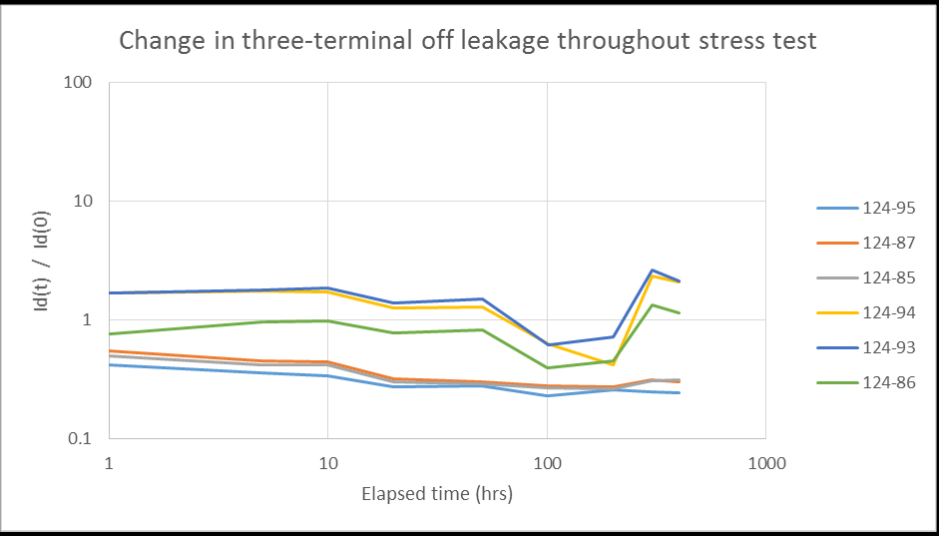
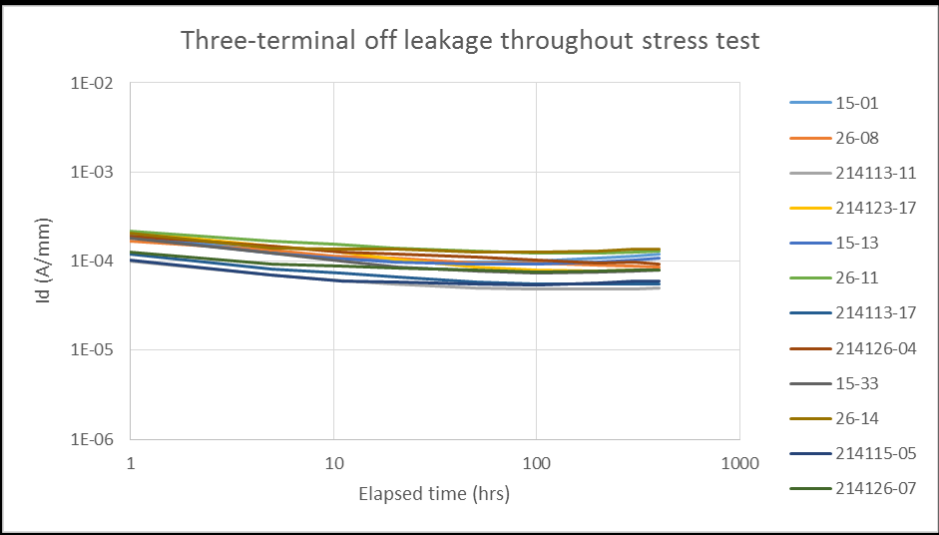


Three-terminal off leakage



GCS

MACOM

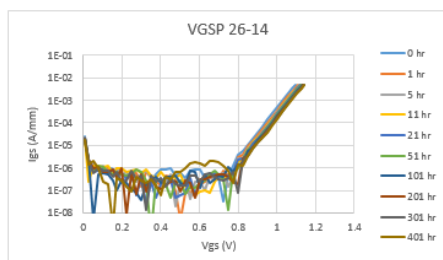
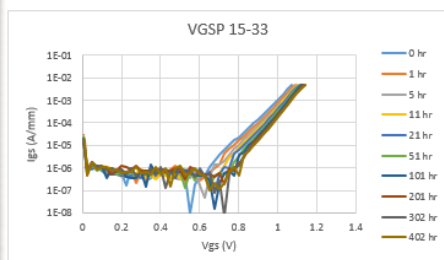
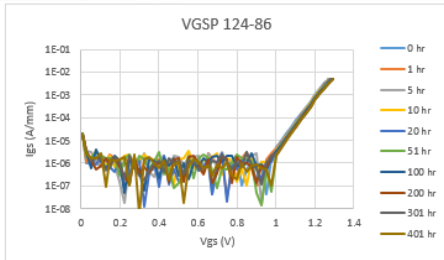
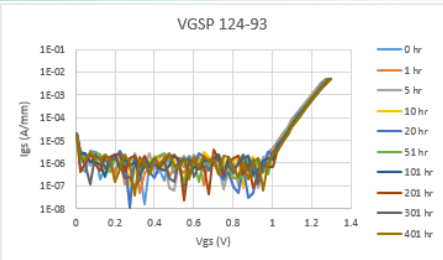
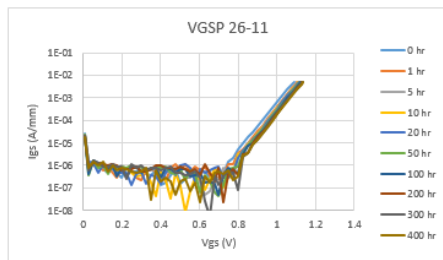
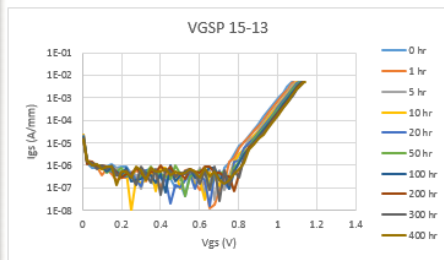
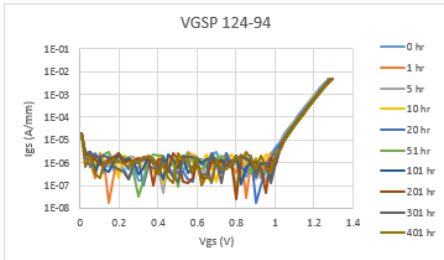
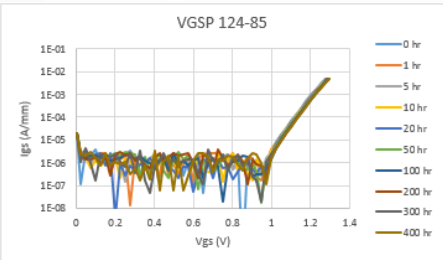
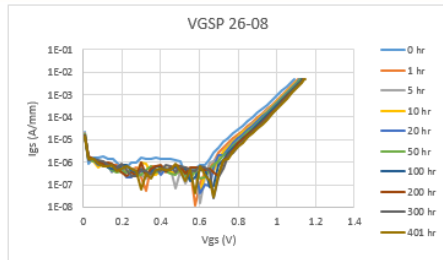
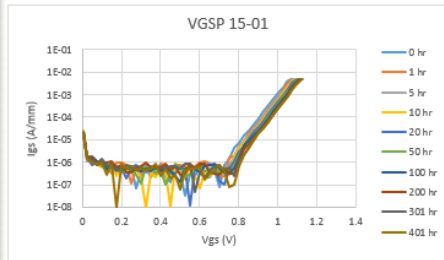
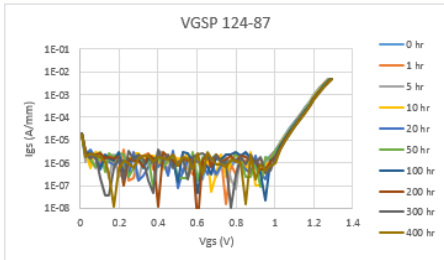
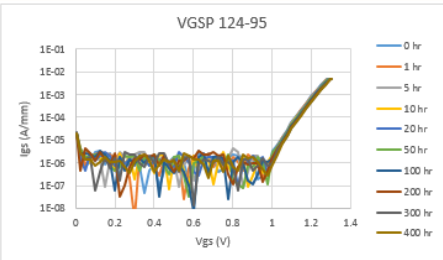


Gate-source diode



GCS

MACOM



No discernible leakage
Stable turn-on voltage

No discernible leakage
Positive shift in turn-on voltage

MACOMTM

Partners from RF to Light

GSi40 Idq Drift Under RF Drive

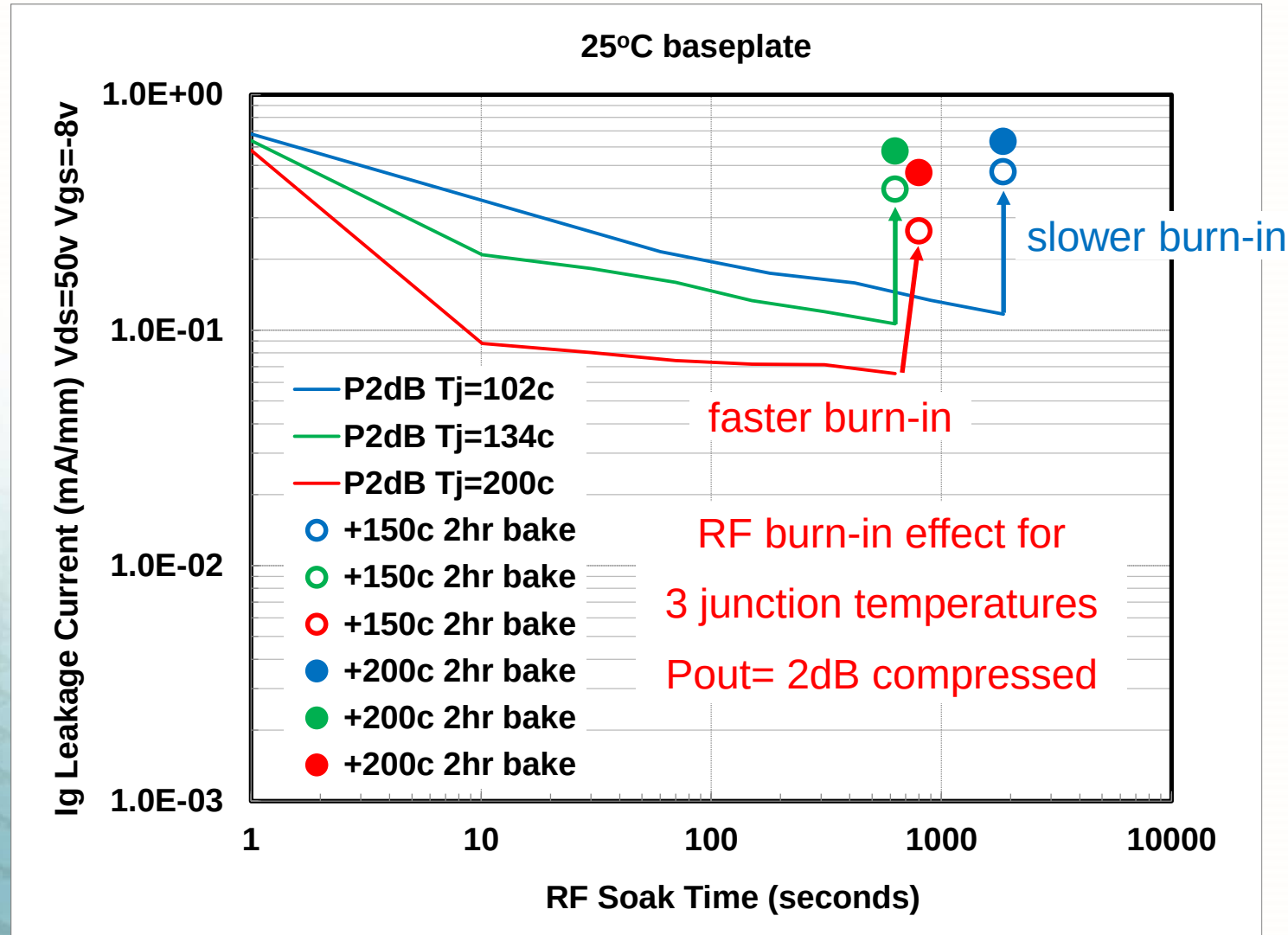


Summary

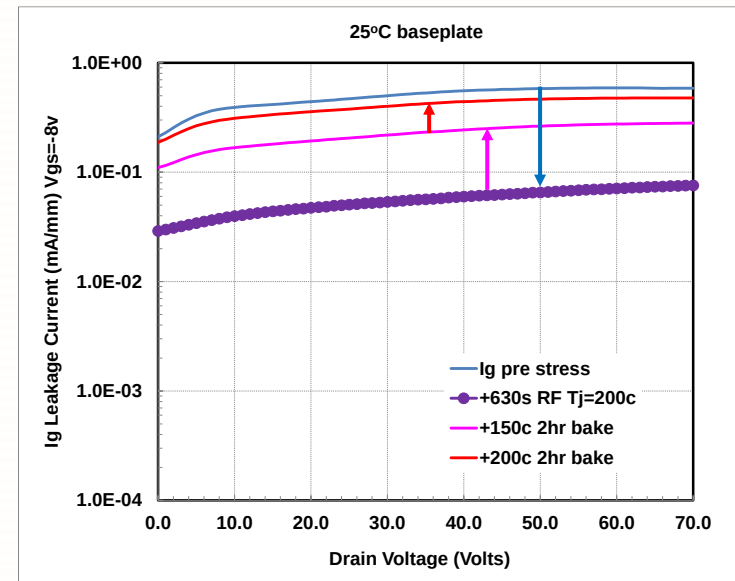
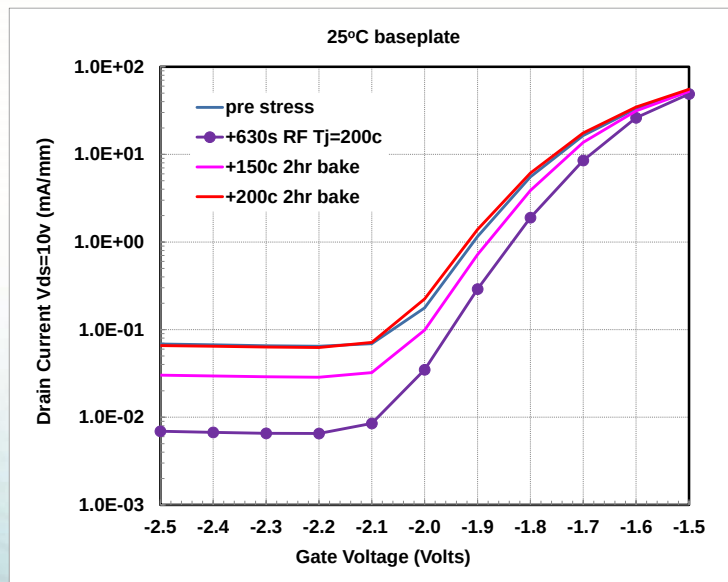


- Lowell GSi40 0.4 μ m GaN on Si
 - The GSi40 GaN on Silicon process exhibits a device burn-in effect under RF drive.
 - We performed a number of experiments looking at the affects of time, temperature, and RF drive on all DC device characteristics.
 - We have determined that the device characteristics (V_{po} , leakage, V_{fwd} , etc.) all change under RF drive and temperature, but fully recover after a 150-200°C bake.
 - This means the burn-in effect is trap related.
- Future Work:
 - Further validate these findings and continue testing & charactization
 - Root cause investigation under way and we believe this phenomena can be mitigated
 - Report on reproducibility and constancy of this device behavior.

GSi40 3T Leakage Current Burn-in Effect (versus time/temperature/RF)



GSi40 FET Characteristics Burn-in Effect (versus time/temperature/RF)



RF burn-in effect

Pout= 2dB compressed
with post bake response

Agenda

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End of Presentation

Thank you!

